

Orco regulates the circadian activity of pheromone-sensitive olfactory receptor neurons in hawkmoths

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Aditi Vijayan, Mauro Forlino, Yajun Chang, Pablo Rojas, Katrin Schröder, Anna C Schneider , Martin E Garcia, Monika Stengl

Animal Physiology and Center for Interdisciplinary Nanostructure Science and Technology, FB 10, University of Kassel, Kassel, Germany • Theoretical Physics and Center for Interdisciplinary Nanostructure Science and Technology, FB10, University of Kassel, Kassel, Germany

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eLife Assessment

In this manuscript, the authors used in vivo long-term tip recordings of the long trichoid sensilla of male hawkmoths to analyze spontaneous spiking activity indicative of the ORNs' endogenous membrane potential oscillations. The authors combine extracellular electrophysiology of the hawkmoth antennae with computational modeling to predict that Orco receptor neuron (ORN) activity is required for circadian, not ultradian, firing patterns. The work provides **valuable** support for the hypothesis that a posttranslational feedback loop regulates daily and ultradian rhythms in neuronal excitability. Nevertheless, the evidence reported provides only **incomplete** support for their conclusions, especially with regard to the biological implications of their assumption-heavy models.

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Abstract

The mating behavior of nocturnal *Manduca sexta* hawkmoths is under strict temporal control. It is orchestrated via circadian- and ultradian oscillations in sex-pheromone stimuli as social zeitgeber. The extremely sensitive pheromone-detecting olfactory receptor neurons (ORNs) that innervate the long trichoid sensilla on the male's antennae are peripheral circadian clocks. They express the transcriptional-translational feedback loop (TTFL) circadian clockwork, best characterized in *Drosophila melanogaster*. In hawkmoths, it is still unknown whether or how the ORN's TTFL clockwork regulates the daily rhythms in pheromone-sensitivity and in temporal resolution of ultradian pheromone pulses as prerequisites to the temporal regulation of hawkmoth mating behavior.

We hypothesize that, rather than the slow TTFL clock, a more rapidly adaptive post-translational feedback loop (PTFL) clockwork, associated with the ORN's plasma membrane, allows for temporal control of pheromone detection via generation of multiscale endogenous membrane potential oscillations. The potential oscillations of the PTFL clock could rapidly synchronize to oscillations of pheromone stimuli at different timescales, thus enable the

prediction of stimulus patterns as a mechanism for active sensing. With *in vivo* long-term tip recordings of long trichoid sensilla of male hawkmoths, we analyzed the spontaneous spiking activity indicative of the ORNs' endogenous membrane potential oscillations. Consistent with our hypothesis of a multiscale PTFL clock in hawkmoth ORNs, spontaneous spiking was modulated on ultradian and circadian timescales, with maximum activity at night. When we blocked the evolutionarily conserved olfactory receptor coreceptor (Orco), the circadian modulation was abolished but the ultradian frequency modulation of the spontaneous activity remained. Consistent with PTFL control, Orco was not under the circadian control of the TTFL clock. We could replicate the experimental data in a conductance-based computational model of Orco. In this model, Orco conductivity changed as a function of fluctuating 2nd messenger levels. This study demonstrates that a PTFL is sufficient to impose a circadian pattern on ORN sensitivity.

Significance statement

It is generally assumed that all circadian rhythms in an organism are driven by a transcriptional-translational feedback loop (TTFL) clock. In this study, we demonstrate with *in vivo* recordings of hawkmoth pheromone-sensitive olfactory receptor neurons (ORNs) that the olfactory receptor coreceptor (Orco) is the key pacemaker channel for controlling circadian but not ultradian rhythms in spontaneous spiking activity. Since Orco expression is not driven by the TTFL clock, its conductance is rather controlled by a post-translational feedback loop (PTFL) membrane clock. Accordingly, our computational model suggests that periodic changes in the conductivity of an Orco ion channel, which is mediated by cycling levels of cyclic nucleotides, tune ORN sensitivity. This highlights the role of the contribution of posttranslational modifications to the generation of circadian rhythmicity.

Introduction

Biological clocks generate endogenous physiological and behavioral oscillations on multiple timescales that persist in the absence of any external timing cues under constant conditions (Lamont and Amir, 2010 [↗](#)). Such clocks are essential to generate internal rhythms and to permit their synchronization to external zeitgebers at timescales from milliseconds to years. These synchronized endogenous multiscale clocks thus provide both a continuous time frame for physiology and behavior - the organism's "present time" - and enable organisms to predict and respond to both regular, periodic environmental changes (Stengl and Schneider, 2024 [↗](#)). Even though endogenous clocks are widespread in prokaryotes and eukaryotes (Géron et al., 2024 [↗](#); Hall and Rosbash, 1993 [↗](#); Millar, 2016 [↗](#); Thaïs et al., 2014 [↗](#)), current knowledge about how multiscale clocks are linked over widely diverging timescales remains sparse.

Insect circadian clock neurons, which are tightly connected to visual brain circuits, are amongst the best studied biological clocks (Hall and Rosbash, 1993 [↗](#); Nishiitsutsuji-Uwo and Pittendrigh, 1968 [↗](#); Reinhard et al., 2024 [↗](#)). They regulate physiological and behavioral oscillations that underly sleep-wake rhythms tied to the 24 h light-dark cycles on Earth (Dubowy and Sehgal, 2017 [↗](#)). Circadian clock neurons possess a molecular clockwork of transcriptional-translational feedback loops (TTFLs), which generate endogenous oscillations of clock gene mRNA and protein levels with a cycle period of ~24 h (Hardin, 2011 [↗](#)). It is assumed for both insects and mammals that this TTFL comprises the master clockwork that regulates all physiological and behavioral rhythms in the circadian time range through genetically controlled output pathways (Hastings et al., 2020 [↗](#)).

Molecular TTFL clockworks were not only found in central brain clock neurons but also in sensory neurons, such as olfactory receptor neurons (ORNs) in insect antennae, where they play a role in odor-dependent circadian behaviors like feeding, pollination, reproduction, and navigation (Flecke et al., 2010 [↗](#); Flecke and Stengl, 2009 [↗](#); Krishnan et al., 1999 [↗](#); Merlin et al., 2009 [↗](#), 2006 [↗](#); Page and Koelling, 2003 [↗](#); Plautz et al., 1997 [↗](#); Sauman and Reppert, 1998 [↗](#); Schendzielorz et al., 2015 [↗](#); Schuckel et al., 2007 [↗](#); Tanoue et al., 2004 [↗](#); Zhou et al., 2005 [↗](#)). Little is known about how the TTFL clockwork of insect ORNs regulates circadian rhythms in chemosensory sensitivity that underlie the circadian control of these different behaviors.

At night, male nocturnal *Manduca sexta* hawkmoths are very sensitive to the sex pheromone blend that their conspecific females release in an intermittent pattern (Itagaki and Conner, 1988 [↗](#)). The intermittency is a prerequisite to maintain the male's search flight, which is suspended at pulse frequencies >30 Hz (Baker et al., 1985 [↗](#); Baker and Vogt, 1988 [↗](#); Stengl, 2010 [↗](#)). The pheromone pulses are resolved of up to 3 Hz by pheromone-specific ORNs in the antenna's long trichoid sensilla (Lee and Strausfeld, 1990 [↗](#); Marion-Poll and Tobin, 1992 [↗](#)). Pheromone-specificity is provided by odor receptor (OR) subunits in the ciliary membranes of ORNs that heteromerize with the conserved olfactory receptor co-receptor Orco (Nakagawa and Vosshall, 2009 [↗](#); Stengl and Funk, 2013 [↗](#); Wicher and Miazzi, 2021 [↗](#)). While the various roles of Orco are still under debate, it is known that in heterologous expression systems Orco homomers form a slow, leaky, non-specific cation channel, which depolarizes the cells (Jones et al., 2011 [↗](#); Sargsyan et al., 2011 [↗](#); Sato et al., 2008 [↗](#); Wicher et al., 2008 [↗](#)). *In vivo* tip recordings of long trichoid sensilla in *M. sexta* confirmed that Orco acts as spontaneously opening cation channel, acting as a pacemaker channel, because it promotes ultradian membrane potential oscillations that underly the spontaneous action potential activity (Nolte et al., 2013 [↗](#)). These potential oscillations turn the ORNs into ultradian temporal filters that are sensitive to the frequency of the pheromone pulses they encounter. Pheromone pulses that arrive at a similar frequency as the endogenous sub-threshold membrane potential oscillations would synchronize these oscillations and would be more likely to elicit action potential trains. Therefore, this pacemaker channel-dependent mechanism would allow for active sensing if endogenous membrane potential oscillations are in the physiological range necessary for tracking intermittent pheromone pulses.

It is not known whether the Orco-dependent membrane potential oscillations also occur at circadian timescales, and whether or how they are linked to ultradian potential oscillations. One hypothesis is that the molecular TTFL clockwork of ORNs exclusively controls endogenous circadian rhythms of the membrane potential via circadian expression of the pacemaker channel Orco. Alternatively, the plasma membrane of ORNs could by itself constitute an autonomous clock that is entrained by but does not rely on the TTFL clock. Constant levels of ion channels that do not require daily degradation and daily transcription via the TTFL clock, but with modulation of open-time probability by posttranslational modifications, would constitute a more economical and faster adjustable mechanism to generate endogenous membrane potential oscillations (Stengl and Schneider, 2024 [↗](#)). This would facilitate active sensing, the rapid tuning to detect behaviorally relevant zeitgeber signals in the insect's environment.

Here, we performed a combination of experimental and computational modeling studies to investigate whether Orco plays a role in active sensing at the circadian and/or ultradian timescale via generation of endogenous membrane potential oscillations. We hypothesize that Orco is part of a post-transcriptional feedback loop (PTFL) clockwork in the plasma membrane, rather than being TTFL controlled. To capture naturally occurring, physiologically relevant processes in insect ORNs we performed *in vivo* long-term tip recordings of the spontaneous spiking activity of pheromone-sensitive trichoid sensilla of intact but restrained hawkmoth males.

We corroborated the experimental data with a Hodgkin-Huxley (HH) based computational ORN model. In general, the widely used deterministic HH models do not include circadian control of ion channels. Moreover, the highly dynamic, bursting firing patterns of ORNs cannot be described by

the traditional deterministic modeling of neuronal activity. This constitutes a drawback because stochasticity plays a crucial role in shaping firing patterns which are influenced not only by external stimuli but also by the inherent noise involved in channel gating. Therefore, to address the above points realistically, we developed a Langevin formulation of the HH model, which explicitly accounts for the noise introduced by the random opening and closing of ion channels and circadian modulation of the Orco channel conductivity. By integrating channel noise, we could explore how circadian regulation of ionic currents contributes to the variability of firing patterns and how this noise interacts with external stimuli to influence olfactory signal processing.

Our model aims to highlight the advantages of using a stochastic framework for modeling the circadian activity of ORNs. The combined experimental and modeling efforts generate a more complete picture and predictions concerning these autonomous processes in sensory neurons, which are required for future exploration of physiologically relevant anticipatory behaviors that are driven by biological clocks.

Methods

Animals

Nocturnal *M. sexta* hawkmoths were bred and raised from eggs in the rearing facility at the University of Kassel. Males and females were separated during pupal stages. Males were housed isolated from the females to avoid exposure to female pheromones. All experiments were performed with adult males reared in a long-day photoperiod (L:D 17:7 h to prevent diapause) at 25 °C with relative humidity of about 55%. Larvae were fed with an artificial diet modified after Bell and Joachim (1976) [\[1\]](#); adult moths could feed on sugar solution with added vitamins (Roth) *ad libitum* (Riffell et al., 2008 [\[2\]](#)).

Solutions

Ringer compositions were taken from (Pézier et al., 2007 [\[3\]](#)) and contained (in mM): 6.4 KCl, 12 MgCl₂, 1 CaCl₂, 12 NaCl, 10 HEPES, 340 glucose for hemolymph ringer (HLR) at 450 mosmol/kg, and 172 KCl, 3 MgCl₂, 1 CaCl₂, 25 NaCl, 10 HEPES, 22.5 glucose for sensillum lymph ringer (SLR) at 475 mosmol/kg, both at pH 6.5. The Orco antagonist OLC15 (N-(4-butyphenyl)-2-((4-ethyl-5-(2-pyridinyl)-4H-1,2,4-triazol-3-yl)thio)acetamide (Chen and Luetje, 2012 [\[4\]](#)); kindly provided by Dr. D. Wicher, Max Planck Institute for Chemical Ecology, Jena, Germany) was dissolved in DMSO to a concentration of 100 mM (stock solution stored at 4-8 °C) before being diluted in SLR to a final concentration of 50 μM immediately before the start of the recording. DMSO percentage in the final solution was 0.05%.

Electrophysiology

The spontaneous electrical activity of the unstimulated pheromone-sensitive ORNs in a single long trichoid sensillum was recorded extracellularly as *in vivo* tip recording (Kaissling et al., 1987 [\[5\]](#)). 1-2-d-old male *M. sexta* that were raised in isolation from females and pheromone exposure were fixed in a custom-made Teflon holder with adhesive tape. The right antenna used for recordings was oriented with its dorsal surface facing up and immobilized near the base with dental wax (Boxing wax strips, KerrHawe SA). Electrodes (Ag/AgCl wire in a glass pipette filled with SLR for the recording electrode and HLR for the reference electrode) were pulled (DMZ-Universal Puller, Zeitz Instruments) from borosilicate capillaries (OD:1.56mm, ID: 1.17mm, without filament; Science Products) to a tip diameter of about 2 μm. After cutting off 20 distal annuli of the antenna with micro scissors, the reference electrode was inserted into the antennal lumen up to the second annulus from the cut end. The cut was covered with electrode gel (GE Medical Systems Information Technology) to prevent desiccation. Electrodes were moved with manual micromanipulators (MMJ, Märzhäuser Wetzlar; and Leitz, Leica Microsystems) or with motorized

micromanipulators (Microstar, Scientifica) with a joystick controller (Scientifica). Under microscopic control, some tips of the long trichoid sensilla of the distal second annulus were cut off with sharpened Dumont forceps, and the recording electrode slipped over a single sensillum of an upper sensillar row.

ORN activity was amplified 200-fold (custom built amplifier with $10^{12} \Omega$ input impedance, or ELC-01 MX in combination with DPA-2FX, npI, or EXT 10-2F, npI), low pass filtered at 1.3 kHz, digitized at 20 kHz with a Digidata (1550A or 1550B, Molecular Devices), and saved for offline analysis with Clampex 10 (pCLAMP suite, Molecular Devices).

All tip recording experiments were performed at room temperature (21-27 °C) and a relative humidity of 35-60%. We recorded ORN activity either in long-day conditions (17:7 LD) or constant darkness (DD) to examine the free-running peripheral ORN clock. For LD experiments, the light regime continued as in the rearing facility during the experiment. For DD and OLC15 experiments, the lights were turned off at the beginning of the recording, i.e. all animals were exposed to the same zeitgeber until the beginning of the experiment. To examine the effect of Orco on the spontaneous action potential (AP) activity pattern, we used the recording electrode to infuse OLC15 in DD conditions.

Data analysis

To characterize the spontaneous ORN spiking activity, we first detected spikes with a self-written program in Python. Using the `scipy.signal` package and `find_peaks` function, we applied a Bessel filter to the raw voltage trace to facilitate spike detection within the following conditions: spike full width (3-15 ms), threshold search for amplitude of spike (0.5-2.5 mV) and threshold prominence (0.1). Two or more consecutive spikes with intervals ≤ 50 ms were considered a burst (Dolzer et al., 2001). We then calculated the following attributes in bins of either 10 min or 1 h for further analysis: instantaneous spike frequency (1 / inter-spike interval (ISI)), inter-burst interval (IBI, time between two consecutive bursts with no single spikes in between), inter-event interval (IEI, time between two events, i.e. two consecutive single spikes, two consecutive bursts, a burst and a consecutive spike, and vice versa), burst duration (time from first to last spike in a burst), percentage of all spikes in the experiment that are members of burst, and mean number of spikes per burst (Figure 1). All statistics were done with those binned values.

Hawkmoth behavioral and physiological rhythms desynchronize across the population in the absence of zeitgebers like pheromone stimulation and light-dark cycles. Therefore, it was necessary to phase align the recordings obtained in DD and OLC15. To achieve this, we used an optimization procedure to calculate the optimal temporal lags for each dataset, aiming to maximize the overall cross-correlation between the calculated binned spike frequency series. The optimization was performed using simulated annealing, which provided a set of time shifts (lags) for each series. These lags were applied to shift the frequency data in a manner that maximized the synchronization between the different subjects' data, resulting in a "subjective ZT" with subjective ZT 0 at the first maximum of the binned ISI for each individual. The objective function used for optimization was the negative sum of pairwise correlations between all series after applying the shifts. The resulting alignment allowed us to compare the frequency patterns across the different conditions.

Once the data were aligned, we computed the mean frequency across all subjects for each bin and each condition. Then, the first high activity period was identified by finding the maximum in the mean frequency series. Subsequently, we identified the second high activity period as the time 24 hours after the first one and the first low activity period was found by locating the minimum in the mean frequency series between the two high activity periods.

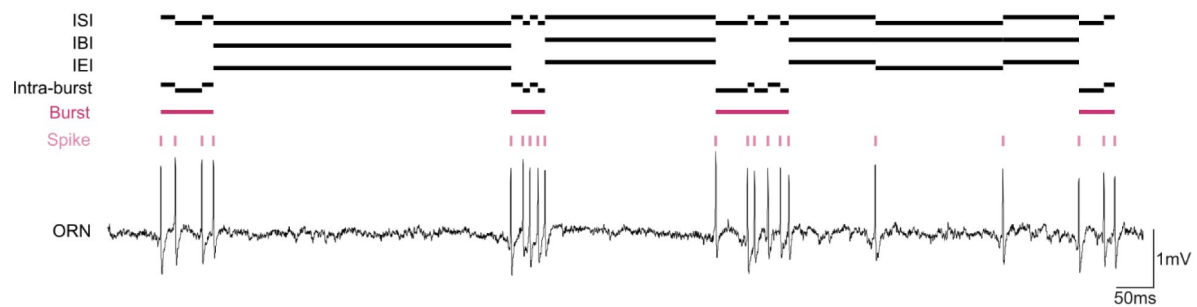


Figure 1

Attributes of pheromone-sensitive ORN spiking patterns.

Spikes were identified by their prominence in the raw ORN recording. The inter-spike interval (ISI) was defined as the time from one spike to the next. A train with ≥ 2 spikes with $ISI \leq 50$ ms was considered a burst. The intra-burst interval was the ISIs within a burst, the inter-burst interval (IBI) the ISIs between two bursts, omitting any individual spikes in between. The inter-event interval (IEI) included the ISIs between bursts and individual spikes.

Besides the mean frequency, we also searched for significant oscillations in IBI, IEI, burst duration, percentage of spikes that are part of burst, and mean number of spikes per burst. For this we found the maximum value of these statistics within a 3 h range centered in the first low activity period and compared it with the values at the second high activity period.

Continuous wavelet and Fourier analysis were performed using PyBoat (Mönke et al., 2020 [↗](#)) to examine the time series of neuronal firing frequency. The continuous wavelet method allows us to detect and characterize rhythmic patterns within the data by identifying frequency components that change over time. Specifically, we used wavelet analysis to determine whether the neuron exhibits circadian and ultradian rhythms. Unlike traditional Fourier analysis, which provides an overall frequency composition but assumes stationarity, wavelet analysis is particularly useful for biological signals because it can reveal how these frequencies evolve dynamically. By applying this approach, we could assess not only the presence of ultradian rhythms but also whether their strength or occurrence fluctuated in a circadian manner, indicating modulation by the 24-hour cycle.

Real-time quantitative polymerase chain reaction (qPCR)

We followed the qPCR methodology detailed in (Schneider et al., 2025 [↗](#)). Briefly, male moths were isolated and cultured until the second day after eclosion, and the antennae from four individuals were collected at each zeitgeber time (ZT) 1, ZT9, and ZT17 using liquid nitrogen, with three biological replicates per time point. Total RNA was extracted using TRIzol® Reagent (Invitrogen) and quantified for concentration and purity with the NanoDrop ND-1000 spectrophotometer (Thermo Fisher Scientific). First-strand complementary DNA (cDNA) was synthesized using the PrimeScript™ RT reagent Kit with gDNA Erase (Takara Bio Inc.). Primers (F: AAGCACGTGGTCAGATTGGT, R: GTAAGCCAGGAGCGTCAGAG) for *Orco* (Gene ID: LOC115452348) were designed using the Primer-BLAST online program of NCBI. The qPCR was conducted following the protocol of TB Green® Premix Ex Taq™ II (Tli RNase H Plus) (Takara Bio Inc.), with the program: 95 °C for 30 s, followed by 40 cycles of 95 °C for 5 s, and 60 °C for 30 s. Relative expression levels of *Orco* were calculated using the $2^{-\Delta\Delta C_t}$ method (Livak and Schmittgen, 2001 [↗](#)).

Statistics

All statistics were done with self-written python codes, SigmaPlot 12 (Systat Software), or JASP (version 0.19.03, University of Amsterdam) with a significance level $\alpha = 0.05$. Unless noted otherwise, data were checked for normal distribution with Anderson-Darling or Shapiro-Wilk tests and for equal variance with Levene's test. If one of these assumptions failed, we used non-parametric tests, otherwise we used parametric tests.

To assess the statistical significance of observed differences in neuronal activity of the aligned data between ZTs, we applied the Wilcoxon signed-rank test to pairwise comparisons between the first low activity period and the subsequent high activity period and discarded the transient high activity at the beginning of most recordings.

For other data, we used an ANOVA on ranks with either Tukey's or Dunn's post-hoc test for multiple comparisons as indicated.

ORN Model

In this study, we introduce a single-compartment Hodgkin-Huxley (HH) type model of the spontaneous activity of *M. sexta* ORNs. The model contains 5 sets of ion channels that were previously identified in electrophysiological recordings (Dolzer et al., 2021 [↗](#)): the fast sodium (Na^+) and potassium (K^+) channels responsible for the generation of spikes, low voltage-activated calcium channels (LVA) responsible for the initiation of bursts, calcium-gated potassium channels (BK) responsible for the termination of bursts, and the Orco channel, which is a voltage-

independent non-specific, cAMP-gated channel. A schematic of the model is provided in **Figure 2A** [↗](#). We modeled Orco as a ZT-dependent conductance to represent the effect of the oscillations in cAMP concentration that were identified in these cells (Schendzielorz et al., 2014 [↗](#)) (**Figure 2B, C** [↗](#)). For simplicity we chose to use a sine function to directly represent the oscillations in Orco conductivity, but similar results can be achieved by modeling it as an output of a Goodwin model of the circadian clock via cAMP concentration.

Each type of ion channel possesses one or more gates that can open and close individually with transition rates α and β for opening and closing, respectively, of activation gates (n , m) and inactivation gates (h). As a result, each possible combination of open and closed gates defines a possible state in which each ion channel can be found. A schematic of the Markov chain depicting all possible states and transitions for each ion channel is shown in **Figure 2D** [↗](#). Each ion channel can conduct current only in the state where all its gates are open.

First, we constructed a deterministic 22-dimensional model to serve as the mean field of the Langevin model following the methodology proposed by Pu and Thomas (2020) [↗](#). Here, one dimension tracks the membrane potential, another the intracellular Ca^{2+} concentration, and the remaining 20 dimensions track the number of channels that are in a specific state of the Markov chains. Considering the conservation of the total number of ion channels, each type of ion channel required one less dimension than the total possible states.

The complete Langevin model adds to the mean field an independent noise source for each transition in the Markov chain. These noise sources are biophysically motivated since the transition between ion channel states is a stochastic process, with the probability of transition determined by the kinetics of each kind of ion channel. The variability in the spiking behavior of a neuron emerges from this non-deterministic gating of individual ion channels. The detailed mathematical description of the model can be found in the Supplementary Material.

Results

We tested the hypothesis that sensory neurons, such as insect ORNs, perform active, anticipatory sensing based upon an endogenous, plastic plasma membrane clock with the pacemaker channel Orco as the core element. We focused on pheromone-sensitive long trichoid sensilla of male *M. sexta* hawkmoth antennae which exhibit daily rhythms in pheromone sensitivity and temporal resolution that are controlled by a circadian clock. We performed minimally invasive *in vivo* tip-recordings over the course of several days to search for daily rhythms in spontaneous spiking activity as a measure of endogenous membrane potential oscillations.

Pheromone-sensitive hawkmoth ORNs exhibit Orco-dependent circadian rhythms in spontaneous spiking activity

The ORNs that innervate the pheromone-sensitive long trichoid sensilla on male *M. sexta* antennae display spontaneous spiking patterns of irregular bursts with interspersed single action potentials in short-term recordings (Dolzer et al., 2001 [↗](#)). In the present study we documented the same pattern in long-term recordings over the course of several days. This enabled us to search for daily and circadian rhythmicity in spontaneous ORN spiking activity in the absence of pheromone stimulation.

The spontaneous spiking patterns revealed clear daily rhythms in 5 of 11 animals under long-day conditions (17:7 h LD) (**Figure 3A** [↗](#)). The maxima in spontaneous activity occurred mostly during the dark phase when nocturnal hawkmoths are active (**Figure 3A** [↗](#) (top), Bi light blue arrow), and minima occurred during the light phase when hawkmoths rest/sleep (**Figure 3A** [↗](#) (bottom), Bi dark blue arrow). High spiking activity in the dark phase had a mean maximum frequency of ~ 3

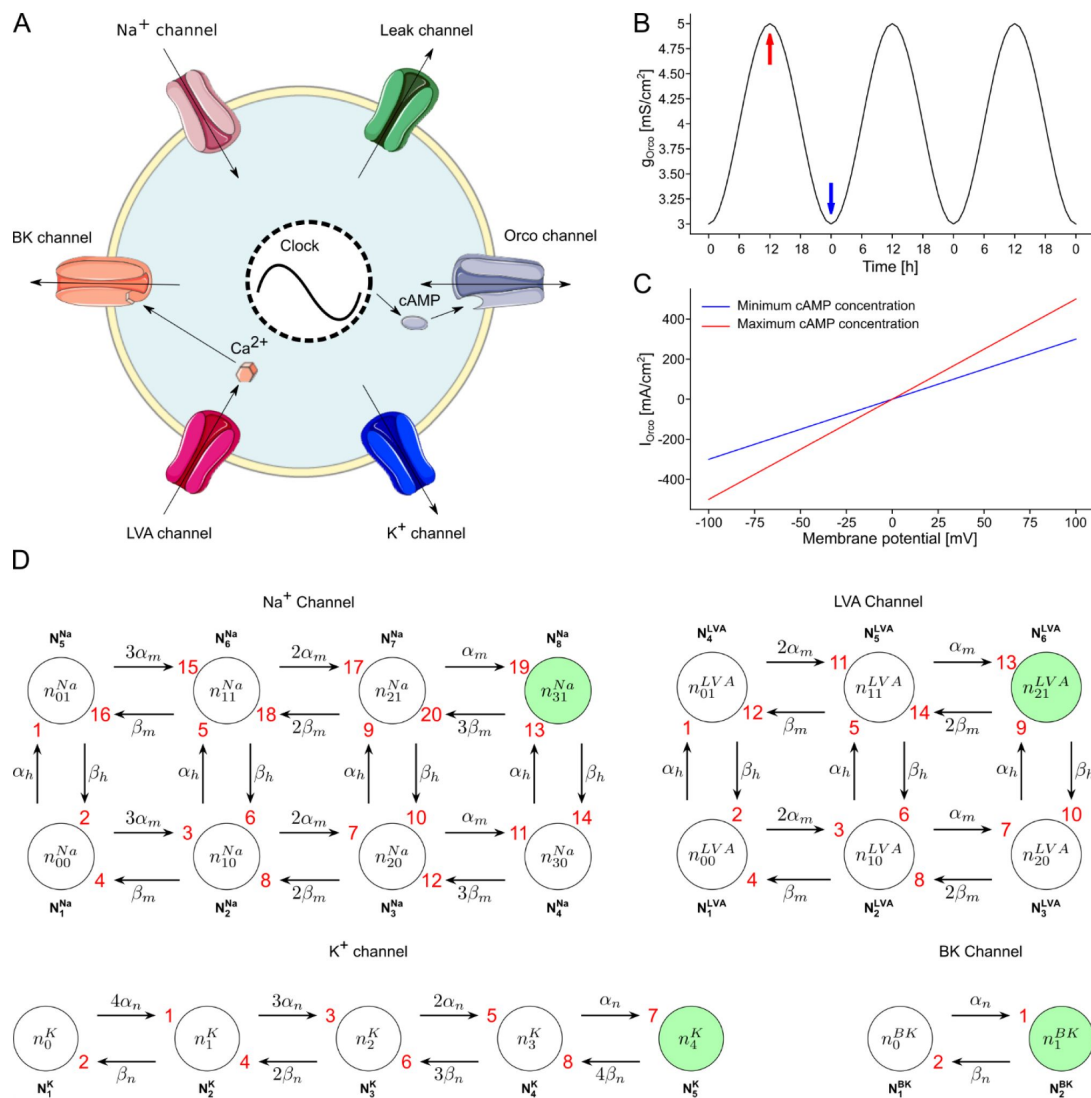


Figure 2

The conductance-based model of a simplified hawkmoth ORN.

(A) Schematic of the model containing a voltage gated sodium channel (Na^+), a voltage gated potassium channel (K^+), a low voltage activated calcium channel (LVA), a calcium and voltage activated potassium channel (BK), a leak channel, and a cAMP-gated leaky non-specific cation channel (Orco). The Orco channel is represented as a linear conductance whose value oscillates with a sinusoidal shape as a function of the ZT. The circadian rhythm in the Orco conductance is hypothesized to result from a cAMP-dependent increase in channel open-time probability (as found in *Drosophila* Orco: (Getahun et al., 2013)) via a circadian oscillation of antennal cAMP concentrations (Schendzielorz et al., 2015). This oscillation is driven by the circadian clock and serves as an input; the internal mechanism of the clock is not modeled explicitly. Arrows indicate the flow of ions through the channels. **(B)** Parametrized circadian oscillation in Orco conductivity, introduced as an external input to our model, representing the gating of Orco by cAMP whose concentration oscillates on a circadian timescale. The times of minimum and maximum cAMP concentration are indicated by colored arrows. **(C)** Parametrized IV curve of the Orco channel showing a linear behavior with the slope depending on the cAMP concentration. Line colors correspond to the arrows in panel B. **(D)** Markov-chain representing all possible states and transitions for each type of ion channel included in the model. The per capita transition rates (α and β) depend on membrane potential and in the case of the BK channel also on the intracellular Ca^{2+} -concentration. The "N" state vectors contain the population of channels in each state "n". Subscripts of "n" indicate how many activation and inactivation gates are in the open state. Directed edges are numbered in red. The only conducting state of each ion channel, representing the condition where all its gates are open, is shown in green. Panel A was drawn using pictures from Servier Medical Art. Servier Medical Art by Servier is licensed under a Creative Commons Attribution 3.0 Unported License.

Hz (theta range: 0.5 – 4 Hz) (**Figure 3Bii** [↗](#)). In contrast, the minimum spiking frequencies during the hawkmoth's resting phase were below 0.5 Hz (**Figure 3A** [↗](#) (bottom)). Since ORN firing patterns had a clear daily rhythm in LD both in individual animals (**Figure 3Bi** [↗](#)) and averaged across animals (**Figure 3Bii, Biii** [↗](#)), we examined whether these rhythms depended on cycling light cues or whether they persisted in constant darkness (DD), as expected for an endogenous, circadian clock-driven rhythm.

In DD, the spontaneous ORN firing pattern remained rhythmic (7 of 10 animals), albeit with less robustness: The single peak of high spiking activity that was observed at the same ZT across several days in LD divided into multiple bouts of high spiking activity in the activity phase in 7 of 10 animals (**Figure 3Ci** [↗](#)). Also, in DD conditions, the high spiking activity phase shifted into the subjective day due to the individual endogenous circadian periods of about 23.5 h $\pm$ 2.8h (21.27 h in **Figure 3Ci** [↗](#)). Thus, averaging the instantaneous spiking activity across all DD animals did not depict clear circadian rhythms (**Figure 3Cii** [↗](#)). Therefore, we phase-aligned the mean frequency of spontaneous activity (see Methods). After alignment, circadian rhythms were exposed across DD animals (**Figure 3Ciii** [↗](#)), comparable to the daily rhythms in LD (cf. **Figure 3 Biii and Ciii** [↗](#)).

Next, we examined whether Orco, a leaky, non-specific cation channel, is the dominant depolarizing pacemaker current of ORNs that drives circadian rhythms in spontaneous spiking activity. Infusion of the Orco antagonist OLC15 into the sensillum lymph obliterated circadian rhythms and attenuated the spontaneous activity in several, but not all experiments (N = 8 of 12) (**Figure 3Di** [↗](#)). This attenuation resulted in a linear decrease in spiking activity over several days (**Figure 3Dii, Diii** [↗](#)). Even after phase-aligning the individual experiments, no circadian rhythm was present in the average across all OLC15-treated animals (**Figure 3Diii** [↗](#)).

Because OLC15 is not membrane-permeable on its own it was infused with low concentrations (0.05%) of DMSO through the recording electrode into the sensillum lymph and, therefore, effectiveness increased over time. Hence, we fit the spiking activity of each animal and in each condition with linear regression lines over the whole recording time to compare the slopes of the fits (**Figure 3E** [↗](#)). In LD, the mean $\pm$ SD slope was 0.004 ± 0.006 Hz/h, indicating that the average spiking activity remained similar throughout the duration of the recording. In DD, the slope was -0.003 ± 0.018 Hz/h and was not different from LD. In contrast, the mean slope during OLC15 treatment was -0.007 ± 0.022 Hz/h, significantly smaller than in LD (ANOVA on ranks with Dunn's post-hoc test; $H(2) = 7.681$, $p = 0.021$). The negative slope corroborated the finding that spiking activity decreased over time in the presence of the Orco blocker.

Activity rhythms between hawkmoths desynchronize and have only weak phase coupling in the absence of pheromone stimulation, even in LD cycles (Ghosh et al., 2024 [↗](#)). Therefore, instead of comparing fixed ZT or circadian time (CT) intervals, we compared different attributes of spontaneous activity between periods of high and low activity, for DD conditions after phase alignment (see Methods; **Figure 4** [↗](#)). Despite the high variability in ORN spiking between individuals, the mean frequency of spontaneous ORN activity was significantly higher during the high activity phase than during low activity in both LD and DD recordings (**Figure 4Ai, Aii** [↗](#)). In LD but not in DD, the frequency of bursting increased significantly during the high activity phase (**Figure 4Bi, Bii** [↗](#)). The percentage of spikes that are part of bursts was less variable and lower during the high activity phase in LD but not significantly different in DD (**Figure 4Ci, Cii** [↗](#)). The mean burst duration was less variable during the high than the low activity phase in LD, but not significantly shorter in any condition (**Figure 4Di, Dii** [↗](#)). Furthermore, both in LD and DD, the mean inter-event frequencies were significantly higher during the high activity phase (**Figure 4Ei, Eii** [↗](#)), indicating a significantly higher overall spontaneous spiking activity during the hawkmoth's activity phase. The mean number of spikes per burst, and intra-burst frequencies did not differ significantly between high or low activity periods in both LD and DD conditions (**Figure 4Fi, Fii, Gi, Gii** [↗](#)). The addition of Orco antagonist OLC15 deleted any significant differences of

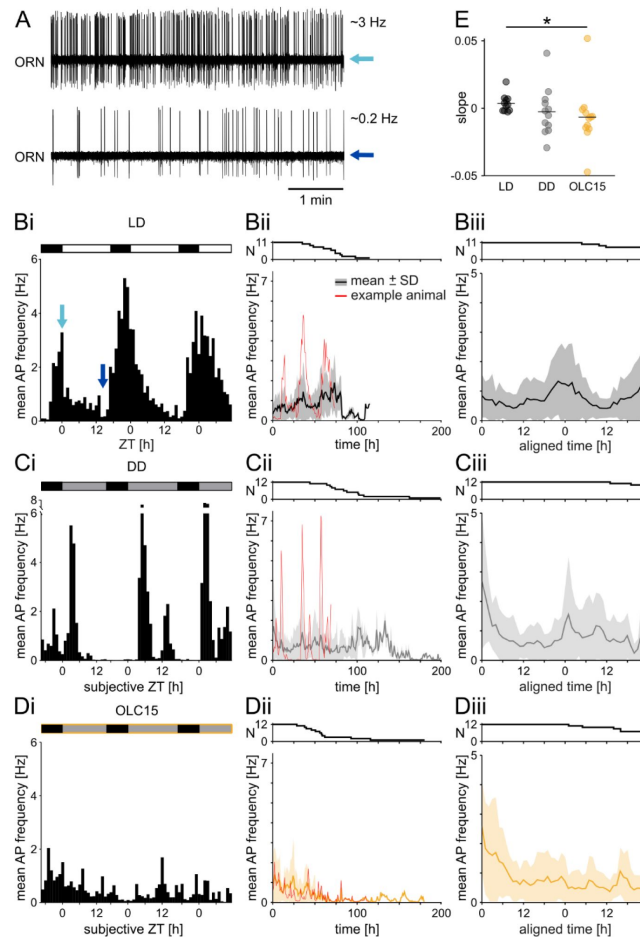


Figure 3

Spontaneous activity of pheromone-sensitive olfactory receptor neurons (ORNs) of male *M. sexta* shows Orco-dependent circadian modulation.

(A) Example tip recordings of the spontaneous activity of one long trichoid sensillum during a period of high (top, average frequency ~ 3 Hz) and low (bottom, average frequency ~ 0.2 Hz) activity. **(Bi)** Spontaneous spike frequency of one long-term ORN recording during 17:7 light-dark cycles (same experiment as in **A**, colored arrows indicate respective zeitgeber times (ZT)), indicated by the black-white bar at the top, increased during each activity phase and was low during the resting phases. Spike frequencies were averaged for each 1 h bin. **(Bii)** Mean spontaneous ORN spike frequency across all LD animals in 1 h bins. Data from the individual in Bi is highlighted in red. Recordings had different lengths, therefore the number of recordings used for the element-wise mean for each 1 h bin decreased with the time since the start of the recording, indicated in the top panel. **(Biii)** Mean spontaneous ORN spike frequency across all LD animals in 1 h bins during the first 48 hours revealed circadian activity. **(Ci)** The spontaneous spike frequency of one long-term ORN recording in constant darkness (DD, the expected times of lights on are represented by grey bars above) exhibited a circadian pattern. The peak activity shifted in DD due to the animal's endogenous, free-running circadian period $\tau = 21.27$ h. **(Cii)** Mean spontaneous ORN spike frequency across all DD animals in 1 h bins. Data from the individual in Ci is highlighted in red. **(Ciii)** Mean spontaneous ORN spike frequency across all DD animals in 1 h bins during the first 48 hours after aligning the time to the first maximum in spontaneous activity (see Methods) revealed circadian activity. **(Di)** The spontaneous spike frequency of one long-term ORN recording in constant darkness with infusion of the Orco antagonist OLC15 (orange frame) dissipated the circadian rhythm of spontaneous activity. **(Dii)** Mean spontaneous ORN spike frequency across all OLC15 animals in 1 h bins. Data from the individual in Di is highlighted in red. In contrast to LD and DD conditions, the spike frequency decreases over time with prolonged exposure to OLC15. **(Diii)** Mean spontaneous ORN spike frequency across all OLC15 animals in 1 h bins (mean $\pm$ SD) during the first 48 hours after aligning the time to the first maximum in spontaneous activity (see Methods). In contrast to LD and DD conditions, the circadian change in spontaneous spiking frequency disappeared. **(E)** The slopes of the linear fits to the binned spiking activity of each individual animal in the three different conditions (see Methods). Each dot indicates the slope for one experiment. The line indicates the mean. The slope for OLC15 is significantly different from control LD.

spontaneous activity found in LD and DD (**Figure 4Aiii-Giii**). In conclusion, the mean frequency of spontaneous spiking and the frequency of bursting expressed daily modulation, and are both most likely controlled via a circadian clock that targets the leak channel Orco.

Orco imposes circadian modulation on the ultradian rhythms in spontaneous ORN activity

After establishing that Orco influences the circadian changes in ORN spiking activity between the animal's resting and activity phase we examined if Orco also influences ORN spiking rhythms on the largely different ultradian timescales in the frequency range between 0.01-1000 Hz. In all conditions, two broad bands of ultradian instantaneous frequencies were evident: one upper band of >10 to ~100 Hz and a lower band between 0.1 to ≤10 Hz (**Figure 5**). The upper band primarily represents the frequencies of spikes within a burst, while the lower band represents single spikes and frequencies between bursts.

In LD, two types of daily/circadian modulation of the ultradian frequencies occurred (**Figure 5A**). The upper higher frequency band was mostly modulated in prevalence (its frequency of occurrence) but not in its frequency range. The lower band was modulated both in prevalence and frequency range, which resulted in a sinusoidal change in the mean of the frequency band over 24 h. The low and high frequency bands appeared to merge during the activity phase around ZT 0 in the animals that showed clear circadian rhythms (N = 5 of 11 in LD), when both bands had maximum prevalence. During the hawkmoth's activity phase, higher frequencies of up to ~10 Hz occurred, while frequencies dropped below 1 Hz during the resting phase.

The circadian modulation of both frequency bands was maintained in constant darkness (**Figure 5B**) but phase-shifted with respect to the projected ZT due to the endogenous circadian period of each individual. Again, this indicates that a clock is modulating the spontaneous ORN spiking activity over the course of a day.

The addition of the Orco antagonist OLC15 deleted any circadian modulation of either frequency band. Interestingly, the instantaneous frequencies in both bands still showed correlated changes in prevalence and merging, but now at ultradian periods only.

Additional Fourier analysis (see Methods) of spontaneous spiking activity over the course of several days revealed faster ultradian and slower infradian rhythms, ranging from 0.6 h to 38 h, in addition to the circadian (24 ± 4 h) rhythm (**Figure 6Ai, Bi, Ci**). Some rhythms (factorials of 24 h, like 12 h, 6 h, etc.) appeared to be harmonics of the circadian rhythm, which were not excluded in our analysis. Most animals expressed ultradian rhythms of 2-4 h in LD as well as DD conditions. With OLC15 the occurrence of circadian rhythmicity decreased from 70% to 33.3 %, while the prevalence of infradian periods increased from 0% to 33.3 % and ultradian periods remained the same. These results were corroborated by continuous wavelet analysis, which provided time-resolved spectral power across different rhythmic periods (**Figure 6Aii, Bii, Cii**). In control LD and DD conditions, wavelet power for ultradian periods increased during times of elevated firing rate and decreased during periods of low activity (**Figure 6Aii and Bii, middle and bottom panels**). This dynamic pattern indicates that ultradian rhythms were not constant over time but were instead modulated by the circadian rhythm, becoming more prominent during circadian peaks in neuronal activity. Such circadian gating of ultradian power was clearly visible in the wavelet spectrograms as alternating bands of higher and lower ultradian power aligned with daily activity cycles. Following Orco blockade with OLC15, this modulation was disrupted (**Figure 6Cii middle panel**): although ultradian components persisted (**Figure 6Cii bottom panel**), they no longer showed systematic variation with the circadian cycle, suggesting a decoupling of ultradian rhythms from circadian control.

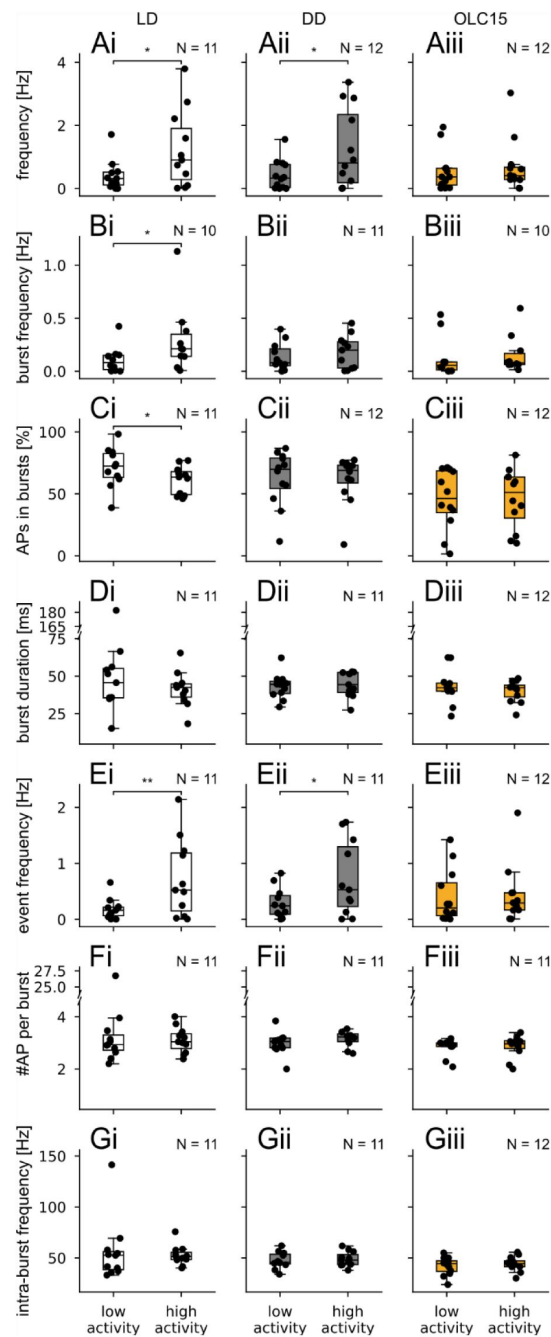


Figure 4

Blocking Orco removed circadian modulation of spontaneous ORN spiking patterns.

Some attributes of the spontaneous ORN activity were significantly different (Wilcoxon signed-rank test, $\alpha = 0.05$) between the time windows of low vs. high activity, mostly in LD (white boxes) and DD (gray boxes) conditions, but never when Orco was blocked with OLC15 (orange boxes). In both LD and DD, the mean spiking frequency was increased during high activity (**Ai**, **Aii**). Only in LD but not in DD, the mean inter-burst intervals (IBI; **Bi**, **Bii**), the relative number of spikes belonging to a burst (**Ci**, **Cii**) and the mean inter-event interval (IEI; **Di**, **Dii**) decreased significantly during the high activity period. Both in LD and DD the mean burst duration (**Ei**, **Eii**), mean number of spikes per burst (**Fi**, **Fii**), and mean intra-burst intervals (**Gi**, **Gii**) did not differ between low vs. high spiking activity. Also, with Orco blocking, the attributes for low vs. high activity did not differ for any of the attributes tested (**Aiii**–**Giii**). In DD and OLC15 conditions, activity phases were aligned as described in the Methods; thus, low activity in DD was at subjective ZT 19 and in OLC15 at subjective ZT 13, high activity was in both cases at subjective ZT 24. ZTs for low and high activity in LD were 10 and 0, respectively.

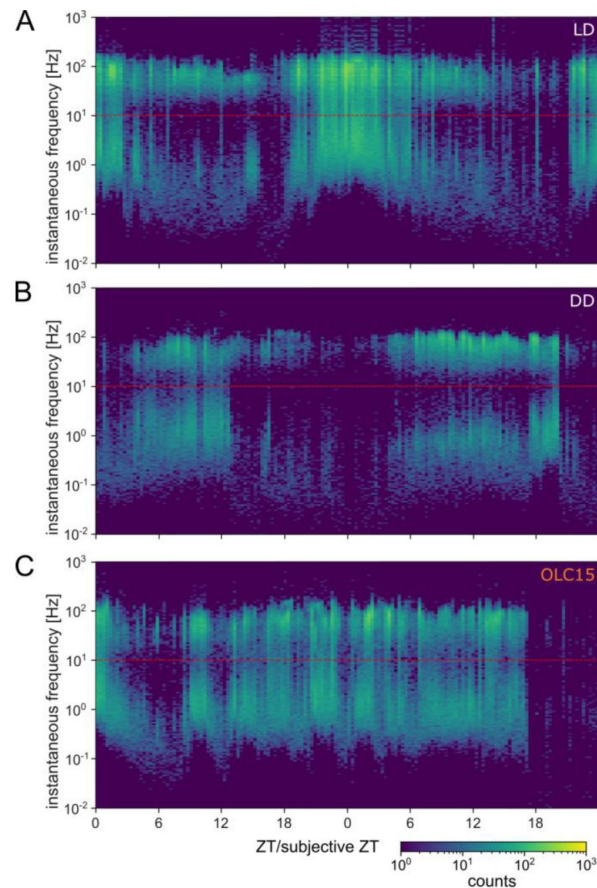


Figure 5

Blocking Orco removed circadian regulation of ultradian rhythms in the spontaneous ORN spiking pattern.

Heat map of the instantaneous frequencies ($1/ISI$) over two consecutive days of long-term tip-recordings of one pheromone-sensitive long trichoid sensillum under different conditions (LD (A), DD (B), OLC15 (C)) in each panel. Pixel color indicates the counts of instantaneous frequencies in that respective bin. The points fall mostly in a band of high frequencies (>10 ~ ~100 Hz) and low frequencies (0.01 ~ ~10 Hz), with the high-frequency band representing the instantaneous frequencies of spikes within a burst and the low-frequency band the instantaneous frequency between bursts. (A, B) The high frequency band indicates daily and circadian modulation of frequency prevalence. In addition, the low frequency band (<0.01 Hz to ~ 10 Hz) also displayed circadian modulation of the frequency composition. (C) Infusion of OLC15 deleted the circadian modulation in both frequency bands and enhanced the ultradian rhythm of frequency prevalence.

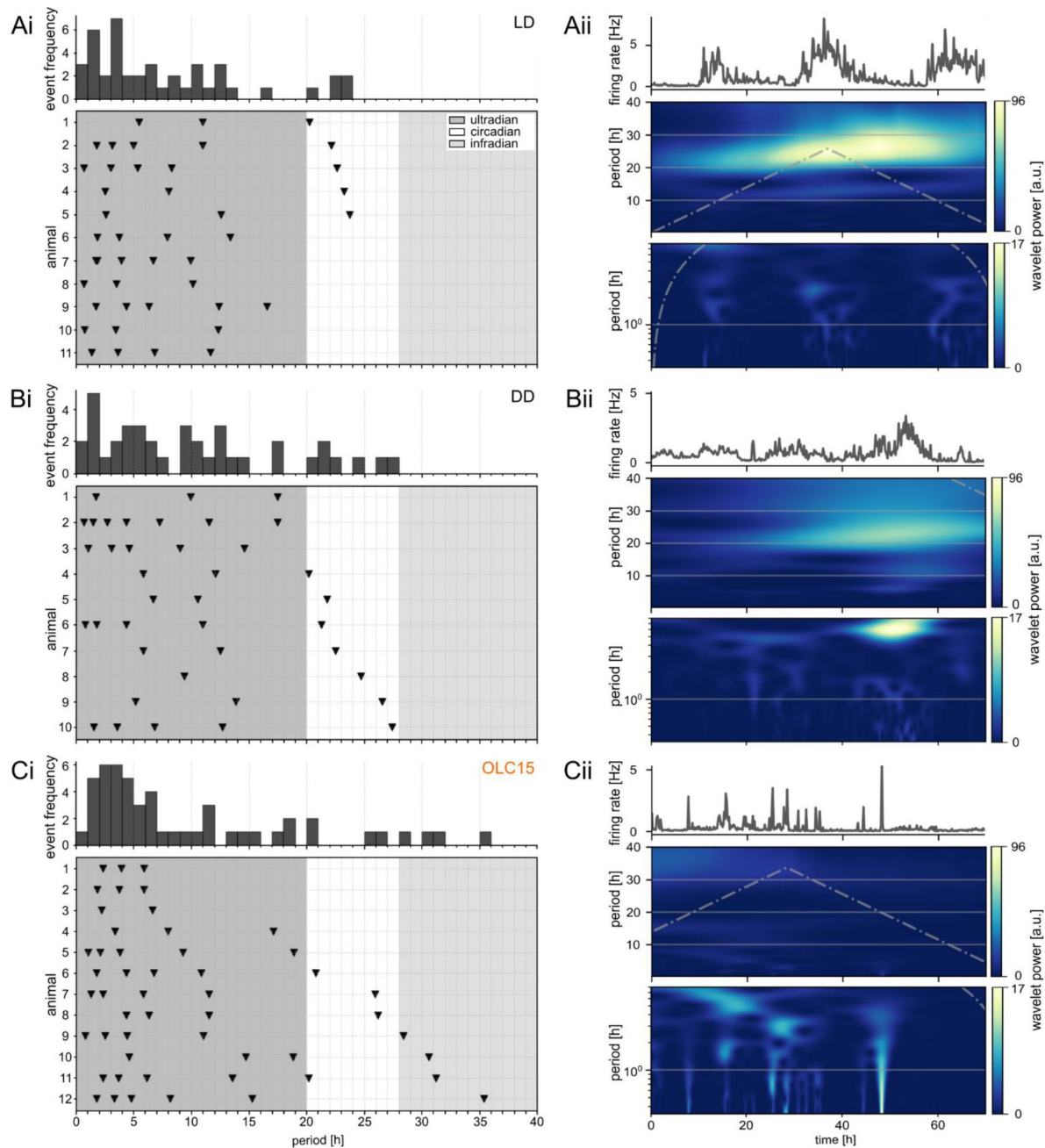


Figure 6

Blocking Orco affected ultradian and infradian frequencies in the spontaneous ORN firing patterns.

The Fourier analysis of spontaneous spiking activity revealed rhythms with ultradian (<20 h; dark grey), circadian (20–28 h; white), and infradian periods (>28 h; light grey) in LD (**Ai**), DD (**Bi**), and OLC15 (**Ci**). Each line represents one animal where each triangular marker is at the local maximum in the frequency spectra obtained for that animal. The histograms above illustrate how often specific periods of spiking activity rhythms occurred averaged over all animals. Circadian rhythms were detected in LD (N = 5 of 11), DD (N = 7 of 10), and OLC15 (N = 4 of 12). Wavelet analysis of LD (**Aii**), DD (**Bii**), and OLC15 (**Cii**) confirmed the occurrence of multiscale periods. Example plots for one animal each, the same animals as in [Figure 5](#). For each panel, the top plot depicts the mean firing frequency, the middle plot the wavelet power in the same period range as panels **i**, and the bottom plot the wavelet power for infradian periods up to 7 h on a log scale to highlight infradian time scales.

Orco expression is not under the control of the molecular TTFL clockwork

So far, our results indicated a prominent role of Orco in the circadian modulation of ORN spiking activity. Since it is the general opinion that the master circadian clock that dominates all physiological and behavioral circadian rhythms functions via transcriptional control we examined whether Orco transcription shows a circadian rhythm, as previously found for clock genes in hawkmoth antennae (Schneider et al., 2025 [DOI](#); Schuckel et al., 2007 [DOI](#)). However, qPCR of whole male hawkmoth antennae revealed no significant circadian rhythm of Orco expression in DD (Figure 7 [DOI](#); ANOVA on ranks, $H(2) = 5.60$, $p = 0.061$). Thus, it is possible that Orco is controlled via a posttranslational circadian clockwork mechanism that is associated directly with the plasma membrane (Stengl and Schneider, 2024 [DOI](#); Stengl and Schröder, 2021 [DOI](#)).

Circadian changes in Orco conductance are sufficient to modulate spiking activity in a computational model of a hawkmoth ORN

Although Orco is a leaky ion channel, its maximum conductivity is modulated by cAMP (Getahun et al., 2013 [DOI](#); Martín et al., 2001 [DOI](#); Sargsyan et al., 2011 [DOI](#); Wicher et al., 2008 [DOI](#)). In hawkmoth antennae, cAMP levels oscillate throughout the day (Dolzer et al., 2021 [DOI](#); Flecke et al., 2010 [DOI](#); Flecke and Stengl, 2009 [DOI](#); Schendzielorz et al., 2014 [DOI](#), 2015 [DOI](#)), which makes cAMP modulation of Orco a putative mechanism for the Orco-dependent circadian modulation of ORN spiking activity, integrating Orco in a circadian PTFL based membrane clockwork (Stengl and Schneider, 2024 [DOI](#)).

To test the hypothesis that Orco is a cAMP-gated, leaky ion channel, we constructed an ORN model (Methods and Supplementary Material) and tuned the parameters to provide a quantitative match with the biological data. The developed model provides a unified description of the experimental observations. For instance, the ORN model yields irregular spiking activity with a bimodal distribution, consisting of both bursts and isolated spikes, successfully reproducing the experimental recordings (Figure 8A [DOI](#)). This suggests a unified mechanism underlying the initiation of both phenomena: If the pacemaker-driven depolarization reaches the Na^+ threshold a firing process begins, and at this point randomness determines if other slower ion channels responsible for bursting are sufficiently activated to sustain a burst of several spikes.

Additionally, the ORN model replicates the distribution of the number of spikes per burst, which follows a decreasing exponential trend in both experimental and simulated results (Figure 8B [DOI](#)). The model also successfully reproduces the circadian modulation of spiking activity: the maximum firing rate occurs when the Orco conductance is maximal, whereas the maximum percentage of spikes that are part of bursts correlates with the lowest Orco conductance, resulting in antiphase oscillations, as were observed in the experimental recordings (Figure 9 [DOI](#)).

The frequency prevalences in the model over several days clustered in two distinct frequency bands: a top band that corresponds to the intra-burst frequencies and a bottom band, corresponding to inter-burst frequencies (Figure 10 [DOI](#)), in agreement with the biological experiments (Figure 5A, B [DOI](#)). When these circadian oscillations in Orco conductance were included, inter-burst frequencies showed significant circadian variations, whereas intra-burst frequencies remained relatively constant, reminiscent of the biological data. Notably, the simulation captured the experimentally observed “merging” effect, where the low frequency band approaches the top band during high-activity periods (Figure 10 [DOI](#)).

This merging of spiking and bursting frequencies suggests that the system approached a bifurcation, transitioning from a bursting regime to tonic spiking activity. Further evidence for the near bifurcation condition of the neurons is the distribution of number of spikes per burst that

Figure 7

Orco is not under control of the TTFL-based molecular circadian clock.

Relative expression levels of *Orco* mRNA from male hawkmoth antennae raised under LD conditions did not differ significantly (ANOVA on ranks, $H(2) = 5.60$, $p = 0.061$) between ZT 1 (end of activity phase), ZT 9 (resting phase), and ZT 17 (beginning of the activity phase). *timeless* mRNA levels from male hawkmoth antenna served as control and were published in Schneider et al. (2025) [\[link\]](#).

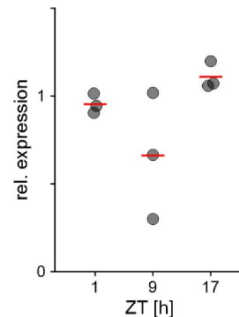
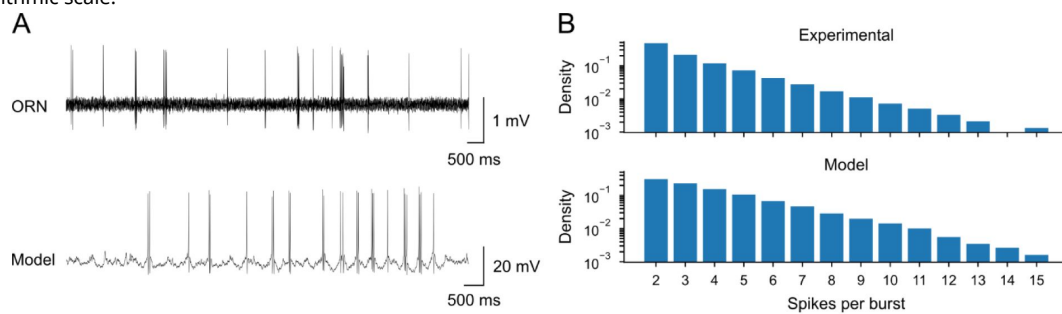


Figure 8

The model adequately reproduces the firing pattern of the biological ORN.

(A) The model activity (bottom trace) displays both isolated spikes and bursts of variable length as in the original recording (top trace). The model represents an intracellular recording whereas the biological recordings were done with extracellular electrodes. **(B)** Semi-logarithmic plot of burst length distribution of an experimental LD recording (top) and simulated model (bottom) of spontaneous action potential activity of pheromone-sensitive neurons. Both the model and the recordings display exponential decrease of the count's density as a function of the burst length, shown as a linear decrease in the logarithmic scale.



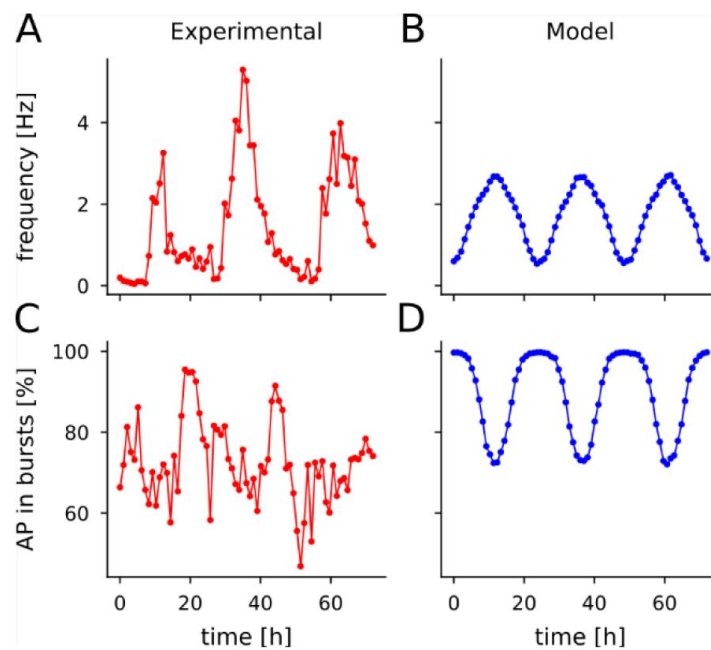


Figure 9

Comparison of the circadian modulation of model and ORN spiking attributes.

Mean spike frequency and percentage of spikes that belong to bursts show anti-phasic circadian oscillations both in the simulated model (left) and in the biological recording (right).

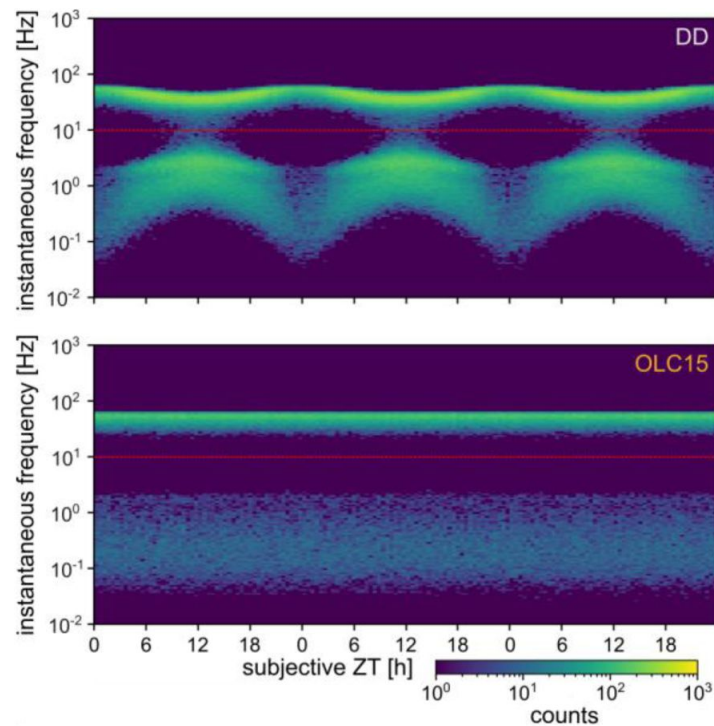


Figure 10

The model captures the Orco-dependent circadian modulation of ultradian spiking rhythms.

Heat map of the simulated model, pixel color indicates the density of instantaneous frequencies in that respective bin. The points can be roughly divided in a band of high frequencies (top, ~50 –80 Hz) and low frequencies (bottom, ~0.1 – 5 Hz), with the high-frequency band representing the instantaneous frequencies of spikes within a burst and the low-frequency band the instantaneous frequency between bursts. **(A)** Simulated spikes with circadian regulation by the Orco channel; the inter-burst frequencies varied widely with circadian rhythmicity while the frequencies within burst remained approximately constant. The simulated results also reproduced the “merging” effect between the two bands when the bottom one approaches the top one. **(B)** Simulated spikes without circadian regulation of Orco had no circadian oscillations.

followed a decreasing exponential trend in contrast to a constant value, or a bounded range of spikes per burst, which would be expected from a standard bursting neuron model (**Figure 8B**). The observation that the circadian modulation of spiking activity requires only one parameter, namely changes in the Orco conductivity, to describe the circadian effects on the ORN spiking pattern indicates the crucial role of Orco channel in circadian control.

Discussion

We searched for a membrane-bound clockwork component in insect ORNs that allows for sensing of regular pheromone patterns to guide mating flights. For this, we recorded interlinked circadian and ultradian frequencies in the spontaneous spiking activity of the pheromone-sensitive long trichoid sensilla of male *Manduca sexta* hawkmoths with *in vivo* long-term tip recordings. We found that Orco, which serves several important functions in insect olfaction (Stengl and Funk, 2013; Wicher and Miazzi, 2021), acts as a pacemaker channel that controls the circadian modulation of the spiking activity without affecting the frequency ranges that are relevant for the primary fast, phasic pheromone response (Dolzer et al., 2003; Nolte et al., 2016; Schneider et al., 2025). While circadian rhythmicity disappeared in the presence of the Orco blocker OLC15, ultradian activity rhythms remained. We confirmed with computational modeling that the circadian modulation of open-time probability of Orco as a leaky pacemaker channel was sufficient to account for our experimental results. Since we observed no evidence for circadian transcriptional control of Orco, our findings are consistent with the hypothesis that the plasma membrane constitutes an endogenous post-translational feedback loop (PTFL) clockwork. In this clockwork, circadian changes in cyclic nucleotide levels (Schendzielorz et al., 2015) modulate Orco conductance rather than transcription and insertion of additional Orco channels in the membrane.

The expression of a pacemaker channel that does not require daily degradation and transcription would constitute an economic and rapidly adjustable mechanism to generate endogenous membrane potential oscillations that are tuned to detect behaviorally relevant zeitgeber signals in the insect's environment (Stengl and Schneider, 2024; Stengl and Schröder, 2021). The present study provides the first evidence for a systems view of timing that involves a PTFL membrane clock in hawkmoth ORNs, instead of the prevailing hierarchical view that assumes all endogenous circadian oscillations are outputs of the genetic TTFL clock (Stengl and Schneider, 2024; Stengl and Schröder, 2021).

Orco acts as a pacemaker channel to drive the circadian rhythm in spontaneous ORN spiking activity

A common motif of spontaneously active neurons is the expression of pacemaker channels that open at hyperpolarized potentials and thus depolarize the neuron (Bose et al., 2014; Cochet-Bissuel et al., 2014; Das et al., 2016; Golowasch et al., 2017; Lüthi and McCormick, 1998; Ratliff et al., 2021; Robinson and Siegelbaum, 2003; Sharma et al., 2023). The resting membrane potential of most neurons lies close to the negative equilibrium potential for potassium ions due to potassium leak channels (Goldstein et al., 2001; Patel and Honoré, 2001; Talley et al., 2001). When the pacemaker-driven depolarization reaches spike thresholds, voltage-gated Na^+ , K^+ and Ca^{2+} channels promote spiking. Voltage- and Ca^{2+} -gated K^+ channels then repolarize the neuron to negative membrane potentials, re-starting the cycle.

The core element of the different frequencies of endogenous oscillations that needs to be identified is the respective pacemaker channel (Stengl and Schneider, 2024). The finding of distinct rhythms in spontaneous spiking activity under constant conditions are indicative of underlying endogenous membrane potential oscillations. Therefore, different sets of voltage- and Ca^{2+} -dependent ion channels with antagonistic effects on the membrane potential appear to

underlie different ultradian oscillation frequencies of spontaneous activity that still need to be further elucidated (Dolzer et al., 2021 [DOI](#); Stengl, 1994 [DOI](#), 1993 [DOI](#); Zufall et al., 1991 [DOI](#)). While this study did not aim to unravel the overall generation of spontaneous activity in the ORNs, we could demonstrate that Orco acts as a circadian pacemaker channel that drives the circadian modulation of the spiking activity.

The constitutively expressed Orco is a core element of a circadian PTFL clockwork in hawkmoth ORNs

In hawkmoth ORNs, at least two types of pacemaker channels exist: the highly conserved, slow, leaky Orco, which we focus on in this study, and an HCN type channel (Butterwick et al., 2018 [DOI](#); Dolzer et al., 2021 [DOI](#); Nolte et al., 2016 [DOI](#), 2013 [DOI](#); Sato et al., 2008 [DOI](#); Stengl and Funk, 2013 [DOI](#); Stengl and Schneider, 2024 [DOI](#); Stengl and Schröder, 2021 [DOI](#); Wicher et al., 2008 [DOI](#); Wicher and Miazzi, 2021 [DOI](#)).

When expressed *in vitro* in heterologous expression systems, *Drosophila* Orco homomers assemble into leaky, non-specific cation channels with a reversal potential around 0 mV, constitutively leaking Na^+ , K^+ , and Ca^{2+} (Sato et al., 2008 [DOI](#); Wicher et al., 2008 [DOI](#); Wicher and Miazzi, 2021 [DOI](#)). In *Drosophila*, Orco is regulated by Ca^{2+} /calmodulin and phosphorylation/de-phosphorylation at multiple sites (Cao et al., 2016 [DOI](#); Getahun et al., 2016 [DOI](#); Guo and Smith, 2017 [DOI](#); Mukunda et al., 2016 [DOI](#), 2014 [DOI](#); Sargsyan et al., 2011 [DOI](#)). In *Drosophila*, the Ca^{2+} /calmodulin binding site controls the localization of the OR-Orco complex in the ciliary membrane of ORNs and depends on previous odor stimulation (Cao et al., 2016 [DOI](#); Guo and Smith, 2022 [DOI](#); Jain et al., 2021 [DOI](#)). After phosphorylation of Orco by protein kinase C (PKC), a cyclic nucleotide binding site becomes available, and subsequent binding of cAMP or cGMP increases the open-time probability (Getahun et al., 2013 [DOI](#); Wicher et al., 2008 [DOI](#)). This indicates two complementary mechanisms by which Orco conductance can be regulated: by controlling the number of Orco channels in the membrane or by modulating the open-time probability of existing channels through posttranslational modifications.

Since blocking Orco in *M. sexta* decreased the spontaneous activity and deleted its circadian modulation, Orco is apparently the main target for modulation of circadian activity in hawkmoth ORNs. Since Orco transcription levels were constant, posttranslational modifications of Orco that modulate the open-time probability provide a mechanistic explanation. Increases in cAMP and cGMP levels sensitize or adapt the ORNs to pheromone, respectively (Dolzer et al., 2021 [DOI](#), 2008 [DOI](#); Flecke et al., 2010 [DOI](#), 2006 [DOI](#); Redkozubov, 2000 [DOI](#); Schendzielorz et al., 2015 [DOI](#), 2012 [DOI](#); Ziegelberger et al., 1990 [DOI](#)). While cAMP levels are under circadian control and peak during the activity phase, cGMP levels oscillate in a daily rhythm and peak during the resting phase in insects (Schendzielorz et al., 2014 [DOI](#), 2015 [DOI](#)). We propose that cAMP and cGMP binding modulate Orco with opposing effects, consistent with the physiological needs of the hawkmoth during its sleep-wake cycle: cAMP binding increases Orco conductance while cGMP binding decreases it.

Our hypothesis that a circadian PTFL membrane clock in hawkmoth ORNs controls circadian activity rhythms predicts that the pacemaker channel Orco is part of a membrane signalosome. The protein complexes of this signalosome would comprise the pacemaker channel that generates the specific circadian frequency of endogenous potential oscillations, linked to oscillations of second messenger cascades that determine physiological setpoints of the circadian sleep-wake cycles. Thus, the circadian cAMP oscillations could result from a PTFL membrane clock comprising of a signalosome that minimally consists of voltage-gated Ca^{2+} -channels, Orco, and membrane-anchored adenylyl cyclases (ACs). Since the membrane potential directly influences intracellular Ca^{2+} levels, the AC activity could be periodically upregulated by the circadian oscillations of the endogenous membrane potential oscillations. Alternatively, AC expression could be under the control of the molecular TTFL clock, which would put Orco under indirect TTFL control. This possibility should be further explored in hawkmoth antennae. Instead of, or in addition to, the

voltage-gated Ca^{2+} channels the signalosome could also work if it contained octopamine receptors that cause circadian rises in cAMP levels through activation of $G_{\alpha s}$, as found in hawkmoths (Dacks et al., 2006 [↗](#); Flecke et al., 2010 [↗](#)). Since octopamine levels in the hemolymph exhibit circadian oscillations and octopamine sets a physiological state of heightened alertness in insects, the circadian PTFL clock in ORNs would synchronize with endogenous physiological rhythms via octopamine receptor activation (Flecke and Stengl, 2009 [↗](#); Schendzielorz et al., 2015 [↗](#)).

Circadian and ultradian rhythms in spontaneous ORN activity resemble the physiologically relevant timescales for active sensing

Moths are more sensitive to pheromone stimulation during their activity phase than at rest, apparently due to antennal circadian clocks (Flecke et al., 2010 [↗](#); Flecke and Stengl, 2009 [↗](#); Merlin et al., 2007 [↗](#), 2006 [↗](#); Schendzielorz et al., 2015 [↗](#); Schuckel et al., 2007 [↗](#); Silvegren et al., 2005 [↗](#)). The greater the distance from the female, the lower the pheromone concentration. The increased pheromone sensitivity during the activity phase enhances the male's chance to detect a calling female.

Pheromone concentration is encoded in the frequency of the first, phasic burst of spikes of the characteristic triphasic spiking sequence of ORNs in response to brief pheromone pulses, which is independent of Orco (Dolzer et al., 2003 [↗](#); Nolte et al., 2013 [↗](#); Schneider et al., 2025 [↗](#)).

However, through the cAMP-dependent increase in Orco conductance during the activity phase, the ORN membrane potential is closer to spike threshold and fewer pheromone molecules are necessary to produce a receptor current that brings the membrane voltage over that threshold. Thus, larger Orco conductance increases the likelihood of an ORN response to pheromone stimulation without changing the typical triphasic spiking pattern of the response.

In the odor transduction of *Drosophila*, OR-Orco activation drives the initial receptor current response. In contrast, Orco is not part of an OR-Orco odor-gated receptor ion channel complex in hawkmoths but controls the spontaneous ORN activity and the late, long-lasting pheromone response (LLPR). The LLPR begins several hundred milliseconds after the initiation of the pheromone-dependent transduction current and correlates with the time course of pheromone-induced rises in cGMP levels (Boekhoff et al., 1993 [↗](#); Nolte et al., 2016 [↗](#), 2013 [↗](#); Stengl, 2010 [↗](#); Stengl et al., 2001 [↗](#); Wicher and Miazzi, 2021 [↗](#); Ziegelberger et al., 1990 [↗](#)). Thus, in hawkmoths, Orco is not part of the primary transduction channel but rather controls membrane potential oscillations and thereby the temporal resolution and general pheromone sensitivity of ORNs. The higher spontaneous burst frequency during the hawkmoth's activity phase improves the resolution of the anticipated pheromone pulses during its mating flight in search for females.

In the mammalian suprachiasmatic nucleus (SCN), the clock that controls the sleep-wake cycle, cAMP levels oscillate at an ultradian frequency, and cAMP and cGMP induce opposing phase shifts in SCN spiking activity (Prosser et al., 1989 [↗](#); Prosser and Gillette, 1991 [↗](#)). In *M. sexta* antennae, cAMP levels express daily oscillations and peak during the activity phase. In contrast, cGMP levels do not oscillate but are elevated by pheromone exposure, which adapts the spiking activity (Boekhoff et al., 1993 [↗](#); Dolzer et al., 2021 [↗](#); Flecke et al., 2006 [↗](#); Schendzielorz et al., 2015 [↗](#); Ziegelberger et al., 1990 [↗](#)). This implies that cAMP and cGMP may have opposing effects on Orco conductivity, with results similar to what is seen in the SCN.

The upregulation of neural activity during wakefulness/alertness seems to be ubiquitous among animals. For example, the many synchronously recorded neurons in human EEG show high frequency oscillations in the beta and gamma ranges during alertness and low frequency oscillations in the delta and theta ranges during rest and sleep. Similarly, the spontaneous spiking activity of hawkmoth ORNs is upregulated during their activity phase, albeit only from below delta to the delta/theta range. The increased spontaneous spiking activity in the theta range during the

activity phase of the hawkmoth matches the maximal temporal resolution of pheromone pulses of up to 10 Hz (Marion-Poll and Tobin, 1992 [↗](#)), which supports a role for Orco as part of a membrane-bound PTFL clockwork that tunes the ORNs for active sensing. Even though oscillations in the field potential of electroantennogram responses to odor stimulation can reach 100 Hz in different insects (Szyszka et al., 2014 [↗](#)), these frequencies most likely do not encode physiological odor responses. Apparently, they reflect the astounding adaptive ability of antennal ORNs to synchronize their endogenous membrane potential oscillations across the antenna to a broad range of stimulation frequencies. Pheromone pulse resolution of about 100 Hz was never reported in single olfactory sensilla recordings, and in behavioral experiments male moths stopped their upwind mate search if pheromone pulses exceeded 30 Hz (Baker et al., 1985 [↗](#); Bau et al., 2005 [↗](#), 2002 [↗](#); Tripathy et al., 2010 [↗](#)).

Circadian oscillations in the ORN membrane potential would be ideally suited to tune sensitivity and temporal resolution of the pheromone-sensitive ORNs to the hawkmoth's rest-activity cycle, synchronizing male and female mating behavior. Further experiments are needed to test our hypothesis that ORN signalosomes are highly plastic endogenous PTFL clocks comprising receptors for circadian and ultradian Zeitgebers that allow to tune into internal physiological and external environmental rhythms as basis for active sensing.

Stochastic ion channel dynamics and circadian modulation of Orco conductance explain modeled ORN firing patterns

Our modeling results provide mechanistic insight into the irregular spontaneous activity of pheromone-sensitive ORNs in *M. sexta*, highlighting how stochastic ion channel dynamics and circadian modulation of ion conductance together shape spiking patterns. By employing a Langevin approximation of the stochastic HH formalism, we demonstrate that the intrinsic noise arising from the random gating of ion channels is sufficient to generate both isolated spikes and bursts of spikes, as observed experimentally. Our model effectively recapitulates the spike distributions found *in vivo* and accounts for the role of channel noise as a critical determinant of ORN firing patterns.

A key contribution of our model is the incorporation of the Orco ion channel as a pacemaker component, with a conductance that varies linearly with cAMP concentration. We show that circadian modulation of Orco conductance alone is sufficient to produce the observed time-of-day dependent changes in spike distribution. This finding suggests that the rhythmic expression of cAMP levels across the circadian cycle may act as a primary driver of diurnal changes in ORN excitability, without requiring additional external inputs or modulation of other ion channels.

Importantly, our results reveal that the same set of ion channels can support the emergence of both single action potentials and complex bursting behaviors purely through the stochastic variability of their gating kinetics. This unifying explanation reduces the need for invoking separate biophysical mechanisms or circuit-level inputs to explain these distinct firing modes.

For simplicity, the circadian variation in cAMP concentration was modeled using a sinusoidal function. While this approximation is sufficient for capturing the qualitative features of the rhythmic modulation, future work could enhance the biological realism of the model by coupling Orco conductance dynamics to a molecular circadian oscillator, such as a Goodwin-type transcriptional-translational feedback loop. Incorporating this layer would allow for a more detailed exploration of how molecular clock components regulate cAMP synthesis and degradation. Additionally, further extensions could include downstream regulatory mechanisms such as phosphorylation and dephosphorylation events mediated by kinases and phosphatases under circadian control, potentially contributing to fine-tuned modulation of ion channel function over the daily cycle.

Together, these findings demonstrate the value of a biophysically grounded, noise-inclusive modeling framework for understanding complex temporal patterns in sensory neuron activity and open the door for future integrative models linking molecular circadian clocks to cellular excitability.

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Author contributions

Conceptualization: AV, MF, MEG, MS; data curation: AV, MF, YC, PR, KS, ACS, MEG, MS; formal analysis: AV, MF, YC, PD, ACS; funding acquisition: MEG, MS; investigation: AV, MF, YC, KS; methodology: MF, PR, MEG, MS; project administration: ACS, MEG, MS; resources: MEG, MS; software: AV, MF, PR, MEG; supervision: MEG, MS; validation: AV, MF, YC, PR, KS, ACS, MEG, MS; visualization: AV, MF, YC, ACS, MS; writing – original draft: AV, MF, ACS, MEG, MS; writing – reviewing & editing: AV, MF, YC, ACS, MEG, MS.

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Supplementary Material

Mathematical Model

The mathematical model is a 22-dimensional system of stochastic differential equations designed to describe the behavior of ion channels and their effects on the membrane potential. I_{LVA} represents a low-voltage-activated calcium current, while I_{Na} and I_K are the transient sodium and potassium currents responsible for action potential generation. I_{BK} is a large-conductance potassium channel that is activated by both a depolarized membrane potential and an increased intracellular calcium concentration, and I_L is the leak current. Additionally, the Orco channel is a leaky, non-specific ion channel gated by cAMP, with the cAMP concentration oscillating in a circadian rhythm. To model the Orco conductance, we used a sinusoidal function of Zeitgeber time (ZT), independent of the membrane potential, as shown in **Figure 2B, C** [↗](#). Each of these ion channels has been reported in vitro in previous studies ([Dolzer et al., 2021](#) [↗](#)).

The deterministic part of the 22D model is given by the mean field of the channel-based Langevin model. This approach was introduced by Fox and Lu's (1994) for a 14D HH model. In their framework, one dimension tracks the membrane potential, while the other 13 dimensions correspond to the fraction of channels in each of the five states in the potassium (K^+) Markov chain and the eight states in the sodium (Na^+) Markov chain. Pu and Thomas (2020) demonstrated that the deterministic 14D model and the original 4D HH model are dynamically equivalent, meaning that each solution of the 4D model corresponds to a solution of the 14D model. Expanding upon Pu and Tomas' methodology to represent our system, we added one dimension to track the calcium concentration, six dimensions for each possible state of the LVA channel, and two dimensions for each possible state of the BK channel. The state vectors for each ion channel are therefore defined as follows: a five-component vector N^K for the K^+ gates, an eight-component vector N^{Na} for the Na^+ gates, a six-component vector N^{LVA} for the LVA gates, and a two-component vector N^{BK} for the BK gates. The state vectors are represented as:

$$N^K = [n_{K0}, n_{K1}, n_{K2}, n_{K3}, n_{K4}]^T \in [0,1]^5 \quad (1)$$

$$N^{Na} = [n_{Na00}, n_{Na10}, n_{Na20}, n_{Na30}, n_{Na01}, n_{Na11}, n_{Na21}, n_{Na31}]^T \in [0,1]^8 \quad (2)$$

$$N^{LVA} = [n_{LVA00}, n_{LVA10}, n_{LVA20}, n_{LVA01}, n_{LVA11}, n_{LVA21}]^T \in [0,1]^6 \quad (3)$$

$$N^{BK} = [n_{BK0}, n_{BK1}]^T \in [0,1]^2 \quad (4)$$

The K^+ channel consists of four independent n (activation) gates, forming a five-vertex channel-state diagram with eight directed edges. The channel conducts a current only when it is in the rightmost state (Figure 2D). Similarly, the Na^+ channel is composed of three identical n gates and one h (inactivation) gate, forming an 8-vertex diagram with 20 directed edges, one of which is conducting. The LVA channel includes two identical n gates and one h gate, resulting in a 6-vertex diagram with 14 directed edges, one of which is conducting. The BK channel consists of a single n gate, forming a 2-vertex diagram with two directed edges, one of which is conducting. The Markov chain for every channel is shown in Figure 2B.

For a sufficiently large number of channels, Schmidt and Thomas (2014) and Schmidt et al. (2018) demonstrated that, under voltage clamp, the equations governing the system can be approximated by a multidimensional Ornstein-Uhlenbeck (OU) process, or Langevin equation, as follows:

$$dN^{ion} = \sum_{k=1}^{k_{max}} \zeta^{ion} \left\{ \alpha_k^{ion}(V) N_{i(k)}^{ion} dt + \sqrt{\epsilon^{ion} \alpha_k^{ion} N_{i(k)}^{ion}} dW_k^{ion} \right\} \quad (5)$$

Here ζ^{ion} is the stoichiometry vector for the k^{th} directed edge. If we write $i(k)$ for the source node and $j(k)$ for the destination node of edge k , then $\zeta^{ion} = e_{j(k)}^{ion} - e_{i(k)}^{ion}$. The voltage-dependent per capita transition rate along the k^{th} edge is $\alpha_k^{ion}(V)$ and $N_{i(k)}^{ion}$ is the fractional occupancy of the source node for the k^{th} transition. The finite-time increment in the Poisson process is approximated by a gaussian process, namely, the increment dW_k^{ion} in a Wiener (Brownian motion) process associated with each directed edge. These independent noise terms are scaled by $\epsilon^{ion} = \frac{1}{N_{Tot}^{ion}}$, where $\frac{1}{N_{Tot}^{ion}}$ represents the total number of that specific ion channel.

The membrane potential is governed by the current balance equation:

$$C \frac{dV}{dt} = \sum_{\chi} I_{\chi} \quad (6)$$

where the set χ contains the six ionic currents included in the model:

$$\chi \in \{LVA, Na, K, BK, L, ORCO\} \quad (7)$$

Each of the six ionic currents included in the model are described as:

$$I_{Na} = -g_{Na} N_8^{Na} (V - V_{Na}) \quad (8)$$

$$I_K = -g_K N_5^K (V - V_K) \quad (9)$$

$$I_{LVA} = -g_{LVA} N_6^{LVA} (V - V_{LVA}) \quad (10)$$

[math11

$$I_{BK} = -g_{BK} N_2^{BK} (V - V_{BK}) \quad (11)$$

$$I_{Orco} = -g_{Orco} (0.8 + 0.2 \sin(\frac{2\pi t}{24h})) (V - V_{Orco}) \quad (12)$$

$$I_L = -g_L (V_L - V) \quad (13)$$

Here, g_χ is the maximal conductance, V is the membrane potential, and V_χ is the reversal potential for each ion current. The parameter values for each current can be found in **Table 1** [↗](#).

Each transition rate $\alpha_k^{ion}(V)$ is calculated in the following way:

$$\alpha_k^{ion}(V) = \frac{p_{j(k)\infty}^{ion}(V)}{\tau_p(V)} \quad (14)$$

where $p_{j(k)\infty}^{ion}(V)$ is the steady state value of the specific gate that switches states in the destination state and $\tau_p(V)$ is the time constant of said gate. The steady state activation values and time constants for the LVA, Na, K, and BK channels were based on Viertel and Borisyuk (2019) [↗](#). The steady state functions for the activation gates are denoted with m_∞^{ion} , and those for the inactivation gates with h_∞^{ion} .

Each gate's steady state activation function, except for the BK channel, takes the form:

$$p_{j\infty}^{ion}(V) = \frac{1.0}{1.0 + \exp\left(\frac{(V - \theta_p)}{\beta_p}\right)} \quad (15)$$

Current	g_x (mS/cm ²)	V_x (mV)
I_L	4.3	-62.5
I_{LVA}	10.0	120
I_{NA}	29.17	45.0
I_K	12.96	-105.0
I_{BK}	9.0	-105.0
I_{ORCO}	5.0	0.0

Table 1

Parameter values for model currents.

The I_{BK} current is dependent on the intracellular calcium concentration and membrane potential as represented by the following equations:

$$m_{\infty}^{BK}(V) = \frac{1.0}{1.0 + \exp\left(\frac{\left(\frac{V - V_1}{2}\right)}{(-15.6)}\right)} \quad (16)$$

$$\frac{V_1}{2} = -20 + 59.2e^{-90[Ca]} + 96.7e^{-470[Ca]} \quad (17)$$

$$\tau_{\alpha} = -(q - 1.0) \frac{(f - 0.2)}{0.8} + 170 \quad (18)$$

$$q = 2.9 + 6.3e^{-360[Ca]} \quad (19)$$

$$s = -25.3 + 107.5e^{-120[Ca]} \quad (20)$$

$$f = \frac{1.0}{10.0 \left(e^{\frac{-(V+100.0-s)}{63.6}} + e^{\frac{(-150.0+(V+100.0-s))}{63.6}} \right)} \quad (21)$$

The intracellular concentration of calcium ions is described by the following equation:

$$\frac{d[Ca]}{dt} = -\gamma \frac{I_{LVA}}{zFd} + \frac{[Ca_0] - [Ca]}{\tau_{Ca}} \quad (22)$$

where Ca_0 is the steady state calcium concentration, γ describes calcium buffering, z is the ionic valence of calcium, F is Faraday's constant and d is the depth in microns at which calcium ions concentrate near the membrane.

The parameters of each steady state activation function as well as the time constant for each gating value can be found in **Table 2** [↗](#).

Variable	θ	β	$\tau(\text{ms})$
m^{LVA}	-30	-4.8916	$\frac{40.0}{\cosh\left(\frac{(V + 37.1)}{9.7832}\right)}$
h^{LVA}	-59.2	11.2326	$\frac{350.0}{\cosh\left(\frac{(V + 59.2)}{22.4652}\right)}$
m^{Na}	-25.0	-6.5	0.2
h^{Na}	-26.0	9.0	$\frac{10.0}{\cosh\left(\frac{(V + 26.0)}{18.0}\right)}$
m^{K}	-26.0	-9.0	$\frac{10.0}{\cosh\left(\frac{(V + 26.0)}{18.0}\right)}$

Table 2

Gating variable parameter values.

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Author information

Aditi Vijayan*

Animal Physiology and Center for Interdisciplinary Nanostructure Science and Technology, FB 10, University of Kassel, Kassel, Germany
ORCID iD: [0000-0002-5198-2222](https://orcid.org/0000-0002-5198-2222)

*Both authors contributed equally

Mauro Forlino*

Theoretical Physics and Center for Interdisciplinary Nanostructure Science and Technology, FB10, University of Kassel, Kassel, Germany
ORCID iD: [0009-0004-0061-494X](https://orcid.org/0009-0004-0061-494X)

*Both authors contributed equally

Yajun Chang

Animal Physiology and Center for Interdisciplinary Nanostructure Science and Technology, FB 10, University of Kassel, Kassel, Germany
ORCID iD: [0000-0001-8290-1897](https://orcid.org/0000-0001-8290-1897)

Pablo Rojas

Theoretical Physics and Center for Interdisciplinary Nanostructure Science and Technology,
FB10, University of Kassel, Kassel, Germany
ORCID iD: [0000-0003-3692-9228](https://orcid.org/0000-0003-3692-9228)

Katrin Schröder

Animal Physiology and Center for Interdisciplinary Nanostructure Science and Technology,
FB 10, University of Kassel, Kassel, Germany

Anna C Schneider

Animal Physiology and Center for Interdisciplinary Nanostructure Science and Technology,
FB 10, University of Kassel, Kassel, Germany
ORCID iD: [0000-0002-1270-836X](https://orcid.org/0000-0002-1270-836X)

For correspondence: anna.c.schneider@uni-kassel.de

Martin E Garcia

Theoretical Physics and Center for Interdisciplinary Nanostructure Science and Technology,
FB10, University of Kassel, Kassel, Germany
ORCID iD: [0000-0003-2418-1902](https://orcid.org/0000-0003-2418-1902)

Monika Stengl

Animal Physiology and Center for Interdisciplinary Nanostructure Science and Technology,
FB 10, University of Kassel, Kassel, Germany
ORCID iD: [0000-0002-7896-872X](https://orcid.org/0000-0002-7896-872X)

Editors

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Joint Public Review:

This manuscript puts forward the provocative idea that a posttranslational feedback loop regulates daily and ultradian rhythms in neuronal excitability. The authors used in vivo long-term tip recordings of the long trichoid sensilla of male hawkmoths to analyze spontaneous spiking activity indicative of the ORNs' endogenous membrane potential oscillations. This firing pattern was disrupted by pharmacological blockade of the Orco receptor. They then use these recordings together with computational modeling to predict that Orco receptor neuron (ORN) activity is required for circadian, not ultradian, firing patterns. Orco did not show a circadian expression pattern in a qPCR experiment, and its conductance was proposed to be regulated by cyclic nucleotide levels. This evidence led the authors to conclude that a post-translational feedback loop (PTFL) clockwork, associated with the ORN plasma membrane, allows for temporal control of pheromone detection via the generation of multi-scale endogenous membrane potential oscillations. The findings will interest researchers in neurophysiology, circadian rhythms, and sensory biology. However, the manuscript has limited experimental evidence to support its central hypothesis and is undermined by several

questionable assumptions that underlie their data analysis and model builds, as well as insufficient biological data, including critical controls to validate and/or fully justify the model the authors are proposing.

Strengths:

The study is notable for its combination of long-term in vivo tip recordings with computational modeling, which is technically challenging and adds weight to the authors' claims. The link between Orco, cyclic nucleotides, and circadian regulation is potentially important for sensory neuroscience, and the modeling framework itself - a stochastic Hodgkin-Huxley formulation that explicitly incorporates channel noise - is a solid and forward-looking contribution. Together, these elements make the study conceptually bold and of clear interest to circadian and olfactory biologists.

Major weaknesses:

At the same time, several limitations temper the conclusions. The pharmacological evidence relies on a single antagonist and concentration, without key controls. The circadian analysis is based on relatively small numbers of neurons, with rhythms detected only in subsets, and the alignment procedure used in constant darkness raises concerns of bias. The molecular evidence is sparse, with only three qPCR timepoints, and the model, while creative, rests on assumptions that are not yet fully supported by in vivo data.

Detailed comments are provided below:

(1) The role for Orco proposed in the authors' model largely stems from the effects seen following the administration of (a single dose) of the Orco antagonist, OLC15. However, this hypothesis is undercut by the lack of adequate pharmacological controls, including a basic multipoint OLC15 dose-response series in addition to the administration of blockers for the other channels that are embedded in their model, but which were ruled out as being involved in the modulation of biological rhythms. In addition, these studies would (ideally) also benefit from the inclusion of the same concentration (series) of an inactive OLC15 analog to better control for off-target effects.

(2) The expression pattern of Orco was assessed using qPCR at only three timepoints. Rhythmic transcripts can easily be missed with such sparse sampling (Hughes et al., 2017). A minimum of six evenly spaced timepoints across a 24-hour cycle would be required to confidently rule out circadian transcriptional regulation. In addition, the use of the timeless mRNA control from another study is not acceptable. Furthermore, qPCR analysis measures transcript abundance, not transcription, as the authors repeatedly state. Transcriptional studies would require nuclear run-off or, more recently, can be done with snRNAseq analysis. Taken together, these concerns undermine the authors' desire to rule out TTFL-based control that directly led them to implicate a PTTF-based model.

(3) The modelling presented is based on Orco as a ZT-dependent conductance tied to the cAMP oscillations that were reported by this group in the cockroach and from the presence and functionality in *Manduca* of homomeric Orco complexes that are devoid of tuning ORs. While these complexes have been generated in cell culture and other heterologous expression systems, as well as presumably exist in vivo in the *Drosophila* empty neuron and other tuning OR mutants, there is no evidence that these complexes exist in wild-type *Manduca* ORNs. While this doesn't necessarily undermine every aspect of their models, the authors should note the presence of Orco/OR complexes rather than Orco homomeric complexes.

(4) Some aspects of the authors' models, most notably the decision to phase align/optimize their DD and OLC15 recordings, are likely to bias their interpretations.

(5) The tip recordings from long trichoid sensilla are critical aspects of this study. These recordings were carried out on upper sensillar tips located on the distal-most second annulus. Since there are approximately 80 annuli on the *Manduca* antennae, it is unclear whether the recordings are representative of the antennal response.

(6) The authors do not provide any data in support of their cAMP/cGMP-based Orco gating, and the PTF model proposed is somewhat disappointing. The model seems to be influenced by their long-held proposal that insect olfactory signaling has a critical metabotropic component involving cyclic nucleotides, PKC, etc, a view that may be influenced by the use of Orco homomeric complexes generated in HEK cells. Nevertheless, structural studies on Orco do not support a cyclic nucleotide binding site, although PKC-based phosphorylation has been implicated in the fine-tuning/adaptation of olfactory signaling.

(7) Because only 5/11 LD and 7/10 DD animals showed daily rhythms, with averages lacking clear daily modulation, the methods are not sufficiently reliable enough to reveal novel underlying mechanisms of circadian rhythm generation. The reported results are therefore not yet reliable or quantifiable. To quantify their results, the authors should apply tests for circadian rhythmicity using methods such as RAIN, JTK CYCLE, MetaCycle, or Echo. The use of FFT and Wavelet is applauded, but these methods do not have tests of significance for rhythms and can be biased when analyzing data in which there could only be 1-3 circadian cycles. Because the conclusions appear to be based on 11-12 neurons that were recorded for 2-4 days, the reader is concerned that the methods are not yet perfected to provide strong evidence for circadian regulation of spontaneous firing of ORNs. The average data (e.g., Figure 3Bii and 3Cii) highlight the apparent lack of daily rhythms. In summary, the results would be more compelling if more than 50% of the recordings had significant circadian amplitudes and with similar periods and phases.

(8) The statement that circadian patterns of ORN firing are lost with the Orco antagonist (OLC15) is not strongly supported. The manuscript should be revised to quantify how Orco changed circadian amplitude in the 12 recorded neurons. Measures of circadian amplitude can avoid confusing/vague statements like Line 394 "low and high frequency bands appeared to merge during the activity phase around ZT 0 in the animals that showed clear circadian rhythms (N = 5 of 11 in LD)". The conclusion that Orco blocks circadian firing appears to be contradicted by Figure 6, which indicates that ~6 of these neurons had circadian periods detected by wavelet. The manuscript would be strengthened with details about the specificity and reproducibility of the Orco antagonist. The authors quantify the gradual decrease in firing with the slope of a linear fit to estimate how the "effectiveness [of OLC15] increased over time." They conclude that the drug "obliterated circadian rhythms and attenuated the spontaneous activity in several, but not all experiments (N = 8 of 12)." The report would be greatly strengthened with corroborating data from additional Orco antagonists and additional doses of OLC15 (the authors use only 50 μ M OLC15).

(9) The manuscript includes several statements that are more speculation than conclusion. For example, there is no evidence for tuning or plasticity in this report. Statements like the following should be removed or addressed with experiments that show changes in odor response specificity or sensitivity: "ORN signalosomes are highly plastic endogenous PTFL clocks comprising receptors for circadian and ultradian Zeitgebers that allow to tune into internal physiological and external environmental rhythms as basis for active sensing." (Discussion Line 622). The paper concludes that (line 380) "mean frequency of spontaneous spiking and the frequency of bursting expressed daily modulation, and are both most likely controlled via a circadian clock that targets the leak channel Orco." This is too bold given the available results.

(10) Because Orco conductance is modulated by cyclic nucleotides, it remains highly plausible that circadian regulation occurs upstream at the level of signaling pathways (e.g., calcium,

calcium-binding proteins, GPCRs, cyclases, phosphodiesterases). The possibility that circadian oscillations of cyclic nucleotides are generated by the canonical TTFL mechanism has not been excluded. In fact, extensive work in *Drosophila* has demonstrated that the TTFL-based molecular clock proteins are required for circadian rhythms in olfaction.

(11) A defining feature of circadian oscillators is the feedback mechanism that generates a time delay (e.g., PERIOD/TIMELESS repressing their own transcription). While the authors describe how cyclic nucleotides can regulate Orco conductance, they do not provide a convincing explanation of how Orco activity could, in turn, feed back into the proposed PTFL to sustain oscillations. For these reasons, the authors should consider:

- (a) Providing a broader discussion of non-TTFL models of circadian rhythms (e.g., redox cycles, post-translational modifications).
- (b) Reassessing Orco expression using a higher-resolution temporal sampling ({greater than or equal to}6 timepoints per 24 h).
- (c) Clarifying or revising the PTFL model to explicitly address how feedback would be achieved. Alternatively, the data may be more consistent with Orco conductance rhythms being regulated by post-translational mechanisms downstream of the canonical TTFL oscillator, as suggested by the *Drosophila* olfactory system literature.

Minor weaknesses:

- (1) The authors should compare the firing patterns of ORN neurons to the bursts, clusters, and packets of retinal efferent spikes reported in Liu JS and Passaglia CL (2011; JBR). By comparing measures in moths to measures in *Limulus*, the authors might be able to address the question: Is the daily firing pattern of ORN neurons likely a conserved feature of circadian control of sensory sensitivity?
- (2) The methods need further details. For example, it is unclear if or how single neuron activity was discriminated and whether the results were compromised by the relatively large environmental fluctuations in temperature (21-27°C), humidity (35-60%), or other cues known to modulate spontaneous firing.

<https://doi.org/10.7554/eLife.108100.1.sa1>

Author response:

Joint Public Review

This manuscript puts forward the provocative idea that a posttranslational feedback loop regulates daily and ultradian rhythms in neuronal excitability. The authors used in vivo long-term tip recordings of the long trichoid sensilla of male hawkmoths to analyze spontaneous spiking activity indicative of the ORNs' endogenous membrane potential oscillations. This firing pattern was disrupted by pharmacological blockade of the Orco receptor. They then use these recordings together with computational modeling to predict that Orco receptor neuron (ORN) activity is required for circadian, not ultradian, firing patterns. Orco did not show a circadian expression pattern in a qPCR experiment, and its conductance was proposed to be regulated by cyclic nucleotide levels. This evidence led the authors to conclude that a post-translational feedback loop (PTFL) clockwork, associated with the ORN plasma membrane, allows for temporal control of pheromone detection via the generation of multi-scale endogenous membrane potential oscillations. The findings will interest researchers in neurophysiology, circadian rhythms, and sensory biology. However, the manuscript has limited experimental evidence to support its central hypothesis and is undermined by several questionable assumptions

that underlie their data analysis and model builds, as well as insufficient biological data, including critical controls to validate and/or fully justify the model the authors are proposing.

We thank the reviewers for their thorough and thoughtful comments and believe that the manuscript will be much stronger once we incorporate the requested changes.

Please note that we used ORN as acronym for “olfactory receptor neuron” throughout the manuscript. ORNs contain odorant receptors (ORs), and in insects these ORs have to associate with the olfactory receptor co-receptor (Orco) in the cilium of the neuron to form functional OR-Orco complexes for odorant detection. Besides this chaperone function, Orco can form homomers with the potential to act as ionic pacemaker channels; a role which we explore in this study.

Strengths:

The study is notable for its combination of long-term in vivo tip recordings with computational modeling, which is technically challenging and adds weight to the authors' claims. The link between Orco, cyclic nucleotides, and circadian regulation is potentially important for sensory neuroscience, and the modeling framework itself - a stochastic Hodgkin-Huxley formulation that explicitly incorporates channel noise - is a solid and forward-looking contribution. Together, these elements make the study conceptually bold and of clear interest to circadian and olfactory biologists.

Major weaknesses:

At the same time, several limitations temper the conclusions. The pharmacological evidence relies on a single antagonist and concentration, without key controls. The circadian analysis is based on relatively small numbers of neurons, with rhythms detected only in subsets, and the alignment procedure used in constant darkness raises concerns of bias. The molecular evidence is sparse, with only three qPCR timepoints, and the model, while creative, rests on assumptions that are not yet fully supported by in vivo data.

Please see our responses to the detailed comments.

Detailed comments are provided below:

(1) The role for Orco proposed in the authors' model largely stems from the effects seen following the administration of (a single dose) of the Orco antagonist, OLC15. However, this hypothesis is undercut by the lack of adequate pharmacological controls, including a basic multipoint OLC15 dose-response series in addition to the administration of blockers for the other channels that are embedded in their model, but which were ruled out as being involved in the modulation of biological rhythms. In addition, these studies would (ideally) also benefit from the inclusion of the same concentration (series) of an inactive OLC15 analog to better control for off-target effects.

The Orco agonist VUAA1 (Jones et al., 2011) binds directly to Orco and increases the channel open time probability. In *M. sexta* hawkmoths, we have already published that VUAA 1 increases the low spontaneous activity of ORNs in a dose-dependent fashion (Nolte et al., 2016). Chen and Luetje (2012) systematically varied the chemical structure of VUAA1 to identify new Orco ligands and discovered 22 Orco Ligand Candidates (OLC) that either activated or inhibited Orco. In their heterologous expression system, Orco was most sensitive to inhibition by OLC15. Based on these results, we published a dose-response curve of OLC15 inhibition (1-100 μ M) using in vivo tip recordings of pheromone-sensitive long trichoid

sensilla of *M. sexta* (Nolte et al., 2016). In that study, we could also demonstrate that OLC15 antagonizes the VUAA1 activation of Orco.

Furthermore, we tested other published Orco antagonists in *in vivo* assays in intact hawkmoths, focusing on amiloride-derived antagonists, because we previously identified an amiloride-sensitive cation channel in hawkmoth ORNs. We found that, in contrast to OLC15, the amilorides HMA and MIA were not Orco-specific but instead affected different targets depending on time-of-day (Nolte et al., 2016). Based on those experiments and the dose-response curves we determined that the Orco agonist VUAA1 (Jones et al., 2011) and the Orco antagonist OLC15 (Chen and Luetje, 2012) worked best in hawkmoth ORNs to target Orco pharmacologically. Based on comparative tests with other published Orco antagonists we settled since then in all further experiments on a dose of 50 μ M OLC15.

We will clarify the Methods section accordingly.

(2) The expression pattern of Orco was assessed using qPCR at only three timepoints. Rhythmic transcripts can easily be missed with such sparse sampling (Hughes et al., 2017). A minimum of six evenly spaced timepoints across a 24-hour cycle would be required to confidently rule out circadian transcriptional regulation. In addition, the use of the timeless mRNA control from another study is not acceptable. Furthermore, qPCR analysis measures transcript abundance, not transcription, as the authors repeatedly state. Transcriptional studies would require nuclear run-off or, more recently, can be done with snRNAseq analysis. Taken together, these concerns undermine the authors' desire to rule out TTFL-based control that directly led them to implicate a PTTF-based model.

We agree with the referees that more time points and a direct comparison between timeless and Orco mRNA levels should be included in this manuscript. We will include these additional qPCR experiments and edit the manuscript to make clear that we measure transcript abundance, but we will not perform snRNAseq analysis due to time- and financial constraints. We are currently working on the transcriptional control of Orco, both during ontogeny and throughout the day but this work in progress is beyond the scope of this manuscript.

(3) The modelling presented is based on Orco as a ZT-dependent conductance tied to the cAMP oscillations that were reported by this group in the cockroach and from the presence and functionality in Manduca of homomeric Orco complexes that are devoid of tuning ORs. While these complexes have been generated in cell culture and other heterologous expression systems, as well as presumably exist in vivo in the Drosophila empty neuron and other tuning OR mutants, there is no evidence that these complexes exist in wild-type Manduca ORNs. While this doesn't necessarily undermine every aspect of their models, the authors should note the presence of Orco/OR complexes rather than Orco homomeric complexes.

Our ELISAs found circadian oscillations in cAMP levels not only in antennae of the Madeira cockroach (Schendzielorz et al., 2014, 2012), but also in hawkmoth antennae (Schendzielorz et al., 2015). We will add the 2015 citation to the Modeling chapter in the Methods section to clarify this.

We agree with the referees that we cannot distinguish between Orco homo- and heteromers in the different compartments of our hawkmoth ORNs. Thus, as the referee suggests, we will add text regarding the presence and localization of OR-Orco heteromers. However, we have indications that Orco homomers could indeed be present in the hawkmoth ORNs. In a heterologous expression system, MsexOrco expression alone was sufficient to increase intracellular Ca^{2+} levels in response to VUAA1 application (Nolte et al., 2013). In

differentiating primary cell cultures of hawkmoth antennae, Orco expression started during a developmental time window where ORNs did not yet express pheromone receptors, and Orco affected spontaneous activity (Nolte et al., 2016). Thus, Orco homomers are present in developing hawkmoth ORNs during a time window where ORNs already express spontaneous activity but cannot heteromerize with pheromone receptors. However, we do not know whether and in what ratio homo- and heteromers of Orco and ORs are present in the respective sensillum compartments of adult hawkmoths (Nolte et al., 2013; Stengl, 1994; Stengl and Hildebrand, 1990).

We will clarify our manuscript accordingly.

(4) Some aspects of the authors' models, most notably the decision to phase align/optimize their DD and OLC15 recordings, are likely to bias their interpretations.

It is consensus that insects display daily and circadian rhythms in pheromone-dependent mating, odor-gated feeding, and egg-laying behavior that phase-locks to environmental rhythms, corresponding with daily/circadian rhythms of sensory neuron physiology (e.g., Merlin et al., 2007; Rymer et al., 2007; Schendzielorz et al., 2015, 2012). However, circadian rhythms can be easily masked by stress, like the disturbances during a very challenging long-term recording experiment over several days. In addition, we observed in our animal raising facility that in LD 17:7 light-dark cycles the originally nocturnal hawkmoths *M. sexta* distribute their activity patterns over the course of the day, finding nocturnal as well as diurnal hawkmoths. Thus, light-dark cycles were not enough to ensure phase-synchronized behavioral rhythms, and it is very likely that the nocturnal hawkmoths rely heavily on pheromone/odor dependent synchronization as also found in other moth species (Ghosh et al., 2024). Here, we used isolated males that were never exposed to the female pheromones so that their circadian activity patterns readily disperse. Therefore, it became necessary in free-running conditions to first determine the respective behavioral rhythm for each animal, and then to phase-align their activity patterns to allow for statistical analysis. Otherwise, circadian differences would average out in a free-running population. As requested by the referees in point (7), we will use additional tests for rhythmicity in each of our recordings and revise the manuscript accordingly.

Assuming that hawkmoths need pheromone presence as additional Zeitgeber, we are currently working on a new set of experiments where we attempt to improve synchronization by exposure to LD cycles and pheromone before DD and OLC15 recordings. We will add these experiments to the manuscript.

(5) The tip recordings from long trichoid sensilla are critical aspects of this study. These recordings were carried out on upper sensillar tips located on the distal-most second annulus. Since there are approximately 80 annuli on the Manduca antennae, it is unclear whether the recordings are representative of the antennal response.

We think the reviewers might have misinterpreted our description of the recording site. In the Methods, we state that we clip off the 20 most distal annuli (leaving a stump of about 60 annuli) and insert the reference electrode into the flagellum up to the second annulus from the cut end, i.e., the recording site is located at 2/3 – 3/4 of the antenna length as seen from the head of the animal. We will make this more clear in the Methods section.

In addition, our lab did show with antibody stainings against Orco that apparently all ORNs that innervate long and short trichoid sensilla along the whole flagellum express the same staining pattern (Nolte et al., 2016). Furthermore, our patch clamp recordings of primary cell cultures of whole male antennae found largely overlapping ion channel populations across ORNs. This would indicate that all ORNs, whether they express pheromone- or general odorant receptors, could potentially share the same Orco-dependent spontaneous activity

rhythms. In our lab, different experimenters from different years that recorded from long trichoid sensilla on different annuli did not detect obvious differences in neither the spontaneous activity nor the pheromone responses (c.f., Dolzer et al., 2003; Gawalek and Stengl, 2018; Schneider et al., 2025). Thus, it is very likely that we are reporting a general encoding mechanism that is not locally restricted along the antennal flagellum.

(5.1) The authors do not provide any data in support of their cAMP/cGMP-based Orco gating...

There are publications supporting cyclic nucleotide gating of Orco in *Drosophila*, but only after previous phosphorylation via protein kinase C (PKC; review: (Wicher and Miazzi, 2021)). Since Orco is very conserved among insect species, it is likely that these PKC and cGMP/cAMP-dependent regulations are present in other insect species. We are currently running thorough tip-recording experiments on the regulation of Orco gating, which are beyond the scope of this manuscript. However, we will add a set of experiments to this manuscript that demonstrates cAMP gating of Orco.

(5.2)... and the PTF model proposed is somewhat disappointing.

For a detailed introduction of our PTF membrane clock hypothesis please see our opinion paper (Stengl and Schneider, 2024).

(5.3) The model seems to be influenced by their long-held proposal that insect olfactory signaling has a critical metabotropic component involving cyclic nucleotides, PKC, etc, a view that may be influenced by the use of Orco homomeric complexes generated in HEK cells.

Indeed, we propose a metabotropic pheromone-transduction cascade, which in moths and cockroaches is based on G-protein-mediated activation of phospholipase C but not on adenylyl cyclase activation. Our hypothesis is not influenced by HEK cell heterologous expression studies of Orco but is supported by our own work comparing in vivo tip recordings of intact hawkmoths with patch clamp experiments on hawkmoth primary cell cultures of olfactory receptor neurons, which are able to respond to their species-specific pheromones in vitro (Schneider et al., 2025; Stengl, 2010; Stengl and Funk, 2013; Wicher and Miazzi, 2021). In addition, a multitude of publications by other laboratories with in vivo and in vitro studies using physiological, genetic, and immunocytochemical assays all support a metabotropic signal transduction cascade in insect olfaction (reviews: Stengl, 2010; Stengl and Funk, 2013; Wicher and Miazzi, 2021). In contrast, the hypothesis suggesting a solely ionotropic pheromone- and general odor-dependent transduction cascade for all insect species is based on very sparse experimental evidence, based primarily on heterologous expression studies such as HEK cells that lack the insect's WT molecular surroundings, and thus, cannot predict OR-Orco function in vivo. Furthermore, the ionotropic hypothesis is heavily based upon the argument that an inverse 7TM receptor cannot couple to G-proteins, which lacks careful backup via biochemical and structural studies. In addition, the ionotropic hypothesis lacks support via carefully performed physiological in vivo studies in different insect species that paid attention to analysis of the distinct kinetic components of ORN's odor/pheromone responses and that employ physiological concentrations and durations of odor/pheromone stimuli (please see our most recent publication by Schneider et al. (2025)).

(5.4) Nevertheless, structural studies on Orco do not support a cyclic nucleotide binding site, although PKC-based phosphorylation has been implicated in the fine-tuning/adaptation of olfactory signaling.

While structural studies did not find evidence for conserved known cyclic nucleotide binding sites on Orco, this does not exclude the presence of so far unknown binding sites, or via sites that fold out only after a specific sequence of previous phosphorylations of the many phosphorylation sites on Orco. Indeed, physiological studies in *Drosophila* presented evidence for cyclic nucleotide dependence of Orco after previous PKC-dependent phosphorylation (Getahun et al., 2013). Our ongoing in vivo experiments in hawkmoths further corroborate a PKC- and cAMP-dependent modulation of Orco. These studies will be published in a follow-up publication.

(6) Because only 5/11 LD and 7/10 DD animals showed daily rhythms, with averages lacking clear daily modulation, the methods are not sufficiently reliable enough to reveal novel underlying mechanisms of circadian rhythm generation. The reported results are therefore not yet reliable or quantifiable. To quantify their results, the authors should apply tests for circadian rhythmicity using methods such as RAIN, JTK CYCLE, MetaCycle, or Echo. The use of FFT and Wavelet is applauded, but these methods do not have tests of significance for rhythms and can be biased when analyzing data in which there could only be 1-3 circadian cycles. Because the conclusions appear to be based on 11-12 neurons that were recorded for 2-4 days, the reader is concerned that the methods are not yet perfected to provide strong evidence for circadian regulation of spontaneous firing of ORNs. The average data (e.g., Figure 3Bii and 3Cii) highlight the apparent lack of daily rhythms. In summary, the results would be more compelling if more than 50% of the recordings had significant circadian amplitudes and with similar periods and phases.

The long-term tip-recordings of intact hawkmoths are very challenging and take a very long time to accomplish, thus, we are very happy that we succeeded in obtaining so many of them (N=34). Since 5/11 LD recordings and 7/10 DD recordings revealed daily/circadian rhythmicity and since many other physiological recordings at different ZTs of different members of our laboratory all revealed ZT-dependent pheromone-transduction we can be certain that the physiology of hawkmoth antennae is under strict circadian control. Please see also our response to (4) above commenting the phase-dispersal of activity rhythms observed in our experiments, as well as in the behavior of hawkmoth males in the mating cage.

Nevertheless, we will follow the advice of the referees to apply additional tests for significance of rhythms in spontaneous activity, and we are thankful for the tests suggested that we were not aware of.

(7) The statement that circadian patterns of ORN firing are lost with the Orco antagonist (OLC15) is not strongly supported. The manuscript should be revised to quantify how Orco changed circadian amplitude in the 12 recorded neurons. Measures of circadian amplitude can avoid confusing/vague statements like Line 394 "low and high frequency bands appeared to merge during the activity phase around ZT 0 in the animals that showed clear circadian rhythms (N = 5 of 11 in LD)". The conclusion that Orco blocks circadian firing appears to be contradicted by Figure 6, which indicates that ~6 of these neurons had circadian periods detected by wavelet. The manuscript would be strengthened with details about the specificity and reproducibility of the Orco antagonist. The authors quantify the gradual decrease in firing with the slope of a linear fit to estimate how the "effectiveness [of OLC15] increased over time." They conclude that the drug "obliterated circadian rhythms and attenuated the spontaneous activity in several, but not all experiments (N = 8 of 12)." The report would be greatly strengthened with corroborating data from additional Orco antagonists and additional doses of OLC15 (the authors use only 50 μ M OLC15).

We will revise our data analysis, according to the valuable suggestions of the referees.

However, based upon our previous studies with other Orco antagonists and different doses of OLC15 (Nolte et al., 2016) we found that 50 μ M OLC15 is the best Orco antagonist dose in *M. sexta* to target Orco-dependent modulation of spontaneous action potential activity of hawkmoth olfactory receptor neurons. Please see also our response to (1).

(8) The manuscript includes several statements that are more speculation than conclusion. For example, there is no evidence for tuning or plasticity in this report. Statements like the following should be removed or addressed with experiments that show changes in odor response specificity or sensitivity: "ORN signalosomes are highly plastic endogenous PTFL clocks comprising receptors for circadian and ultradian Zeitgebers that allow to tune into internal physiological and external environmental rhythms as basis for active sensing." (Discussion Line 622). The paper concludes that (line 380) "mean frequency of spontaneous spiking and the frequency of bursting expressed daily modulation, and are both most likely controlled via a circadian clock that targets the leak channel Orco." This is too bold given the available results.

We will revise the discussion accordingly and clarify which statements are supported via published evidence and which are predictions based upon our novel hypothesis published in our opinion paper (Stengl and Schneider, 2024).

(9.1) Because Orco conductance is modulated by cyclic nucleotides, it remains highly plausible that circadian regulation occurs upstream at the level of signaling pathways (e.g., calcium, calcium-binding proteins, GPCRs, cyclases, phosphodiesterases).

We agree with the referees that it is very likely that there are multiple layers of interconnected feedback cycles that control Orco localization and activity. Our novel hypothesis suggests interlocked TTFL and PTFL control of physiological circadian rhythms, not strictly hierarchical TTFL control, which would require a daily turnover of membrane proteins and transcriptional control via the established TTFL clock in insect ORNs. We currently search for TTFL control at all levels of odor/pheromone transduction using ZT-dependent transcriptomics in combination with qPCR and single nuclear transcriptomics, involving also all the molecules suggested by the referees. These studies are ongoing, are very time- and money-consuming, and are beyond the scope of this manuscript.

*(9.2) The possibility that circadian oscillations of cyclic nucleotides are generated by the canonical TTFL mechanism has not been excluded. In fact, extensive work in *Drosophila* has demonstrated that the TTFL-based molecular clock proteins are required for circadian rhythms in olfaction.*

Our experiments that test circadian TTFL control at different levels of the cAMP transduction cascade in hawkmoth antennae are on the way and are part of another publication. We will revise our discussion accordingly.

The experiments published for TTFL dependent control of *Drosophila* olfaction that we are aware of (Krishnan et al., 1999; Tanoue et al., 2004) do not exclude interlinked PTFL and TTFL clocks. Krishnan et al. (1999) demonstrate that the TTFL clock in antennal olfactory receptor neurons correlates with circadian rhythms in odor responses measured in electroantennogram (EAG) recordings, not in single sensillum recordings as in our experiments. EAG recordings comprise not only voltage responses of the olfactory sensory neurons but also voltage changes generated in non-neuronal antennal cells such as trichogen and tormogen cells that built the transepithelial potential gradient via vATPases that generates the high K^+ concentration in the sensillum lymph (Jain et al., 2024; Klein, 1992; Thurm and Küppers, 1980). In addition, EAG recordings most likely contain responses of

afferent neurons originating from somata in the brain that maintain central control of the antennae. Thus, EAG recordings are difficult to interpret.

(11) A defining feature of circadian oscillators is the feedback mechanism that generates a time delay (e.g., PERIOD/TIMELESS repressing their own transcription). While the authors describe how cyclic nucleotides can regulate Orco conductance, they do not provide a convincing explanation of how Orco activity could, in turn, feed back into the proposed PTFL to sustain oscillations. For these reasons, the authors should consider:

a) Providing a broader discussion of non-TTFL models of circadian rhythms (e.g., redox cycles, post-translational modifications).

We will revise the discussion accordingly.

b) Reassessing Orco expression using a higher-resolution temporal sampling ({greater than or equal to}6 timepoints per 24 h).

We will add those experiments to the revised version of the manuscript (see our response to (2)).

c) Clarifying or revising the PTFL model to explicitly address how feedback would be achieved. Alternatively, the data may be more consistent with Orco conductance rhythms being regulated by post-translational mechanisms downstream of the canonical TTFL oscillator, as suggested by the Drosophila olfactory system literature.

We will revise the manuscript accordingly.

Minor weaknesses:

(1) The authors should compare the firing patterns of ORN neurons to the bursts, clusters, and packets of retinal efferent spikes reported in Liu JS and Passaglia CL (2011; JBR). By comparing measures in moths to measures in Limulus, the authors might be able to address the question: Is the daily firing pattern of ORN neurons likely a conserved feature of circadian control of sensory sensitivity?

We will revise the discussion accordingly.

(2) The methods need further details. For example, it is unclear if or how single neuron activity was discriminated and whether the results were compromised by the relatively large environmental fluctuations in temperature (21-27°C), humidity (35-60%), or other cues known to modulate spontaneous firing.

We will clarify the Methods section.

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